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Conformal Set Prediction for Phage Host Classification

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Abstract

Bacteriophage therapy is emerging as a promising alternative to antibiotics against multi drug resistant bacteria, but its clinical use depends on correctly matching a phage to a bacterial host it can infect. Existing computational host prediction tools return point predictions or uncalibrated scalar scores, with no finite sample guarantee that the true host is among the candidates returned.

This report develops a distribution free conformal prediction framework for phage host classification that closes this gap. We build on the Conformal Set Aggregation (CSA) approach and replaced its convex calibration envelopes, which may become uninformatively large when the class conditional score clusters are well separated. We used three star shaped, non convex alternatives (Radial, Strip, and Collapsed 1D) which are suited to the disjoint geometry of classification tasks. For each geometry we derived a closed form calibration rule that replaces the iterative binary search used in standard CSA with a single sorting pass. We also introduce a class conditional calibration, which can give exact per host coverage guarantees rather than a single marginal one, along with a two stage Gram type to host cascade with a proven bound on compounded miscoverage.

We validated the framework on a real dataset of bacteriophage protein functional scores, and achieved reliable empirical coverage at the target confidence level on both a binary Gram type task and a multiclass host classification task spanning 54 candidate hosts. Gram type pre filtering also reduced average prediction set sizes by over 60% compared to an unfiltered baseline, and with no loss of coverage.

Contents

— 1 —
Introduction

The rapid global emergence of antimicrobial resistance (AMR) represents one of the most critical threats that modern medicine is facing. As conventional antibiotics lose efficiency against multidrug resistant bacterial pathogens, alternative therapeutic strategies have gained widespread attention. Among these is phage therapy, it is the targeted clinical application of bacteriophages (viruses that specifically infect and destroy bacteria) and has emerged as a highly promising solution [1].

Bacteriophages are the most abundant biological entities in the global biosphere, with an estimated population exceeding 10^{31} particles [1]. One of the main determinants of a phage's therapeutic viability is its host range (the spectrum of bacterial strains it can successfully infect), so matching a bacterial infection with an effective phage requires knowing precisely which bacteria each phage can infect and which it cannot. Section 2.1 reviews the biology of phages and of the host-prediction problem in detail.

Traditional laboratory screening methods, such as plaque assays, are labor intensive, costly, and scale poorly against the millions of uncharacterized viral genomes generated by modern high throughput metagenomic sequencing. This necessitates the development of fast and scalable computational tools that take genomic sequences as input, translate them into protein level representations and then predict the host range from these representations directly without requiring any laboratory validation [2].

However, state of the art computational prediction tools suffer from a fundamental limitation. They produce point predictions or uncalibrated scalar scores in $[0,1]$ without any finite sample uncertainty quantification. In high stakes medical applications, point predictions offer no guarantee of reliability: a confidence score reflects only a relative ranking of candidates, not a calibrated probability that the predicted host is correct. The clinical risk runs in both directions, failing to recognize that a phage kills the infecting bacterium can mean missing a life saving treatment, while overlooking that the same phage may also attack beneficial bacteria can cause further damage. Therefore, sound therapeutic decisions require calibrated knowledge of which phages infect which bacteria, and equally of which they do not; point predictions alone are clinically insufficient.

The main objective of this work is to resolve this limitation by constructing an uncertainty aware statistical framework for phage-host prediction using conformal prediction [3, 4], which is a distribution free method that transforms point scoring models into set valued predictions $C(X) \subseteq \mathscr{H}$. More importantly, these prediction sets are mathematically guaranteed to contain the true host label with a user specified confidence level $1-\alpha$ (e.g., 90%), which holds strictly in finite samples under just the assumption of data exchangeability.

Moreover, standard conformal prediction relies on a single score to evaluate each candidate label, whereas our pipeline evaluates phage host compatibility across K distinct functional protein categories simultaneously. Simply averaging these scores into one value loses useful information, while calibrating each score separately breaks the coverage guarantee; the multi dimensional conformal methods that operate directly on score vectors are introduced in Section 2.4.

In this work we build on the Conformal Set Aggregation (CSA) framework [5]. We address three major challenges that arise when applying CSA to classification datasets:

(i) **Non-convex geometry in classification space:** We demonstrate why traditional convex envelopes, developed primarily for continuous regression spaces, are inefficient in classification settings, since the class conditional score vectors occupy disjoint, bounded clusters in $[0,1]^K$. So, we propose non-convex (Radial and Strip) and Collapsed 1D envelope geometries that adapt to cluster topologies, and eliminate unpopulated empty space, which helps us reduce prediction set sizes.

(ii) **Closed form analytical calibration (τ):** We derive exact, closed form analytical expressions for the per point inclusion scaling function $\tau(\mathbf{s})$ across all envelope geometries. This eliminates the computationally expensive numerical binary search required by standard CSA, which reduces the calibration time complexity from $O(|S_2| \cdot N_{\text{iter}})$ to a single forward pass and $O(|S_2| \log |S_2|)$ sort.

(iii) **Structural missingness handling:** Phage score vectors frequently contain entries that are missing by design, since a phage may lack any protein annotated for a given functional category. We develop the envelope constructions and nonconformity transformations to operate natively on these partially observed vectors via an explicit missingness mask, without imputation.

We evaluated our methodology systematically on two tasks of increasing complexity. First, a controlled binary Gram-type classification task ($N = 2$ classes) to validate the geometric envelope constructions, and then on a multi class host-level prediction task ($N = 54$ bacterial genera) using real-world protein language model embeddings.

The remainder of this report is organized as follows: Chapter 2 reviews the biological background of bacteriophages, computational state-of-the-art methods, and the mathematical foundations of split conformal prediction and CSA. Chapter 3 formalizes the prediction tasks, develops the non-convex envelopes, the nonconformity transformation, and the analytical derivation of $\tau(\mathbf{s})$. Chapter 4 details the dataset architecture, protein to phage NaN aware mean pooling operators, and cross split protein cluster contamination filtering. Chapter 5 presents synthetic experiments and empirical evaluation results for both gram-type and host-level classification tasks. Finally, chapter 6 summarises our findings, discusses the biological limitations, and outlines directions for future research.

— 2 —
Background & State of the Art

In this section, we provide the biological, computational and mathematical background and context for the methods developed in this report. We will begin by introducing bacteriophages and the host-prediction problem in Section 2.1, which motivates the need for computational approaches to find a solution. We will then go over the already existing and state-of-the-art computational techniques in Section 2.2, progressing from the classical sequence similarity techniques to the much more efficient and modern protein embeddings techniques. We will also look at the limitations that all the currently existing methods have in common, for example the absence of rigorous distribution-free uncertainty quantification. In the remaining part of the section, 2.3 and 2.4 we will introduce the mathematical framework of conformal prediction and the score-based aggregation method in [5], on which we will build our work.

2.1 Bacteriophages and Host-Prediction Problem

Bacteriophages, commonly referred to as phages, are viruses that specifically infect and replicate within bacteria. As discussed in Chapter 1, the growing threat of antimicrobial resistance and the development of phage therapy as an alternative to traditional antibiotics [1] have renewed interest in the study of phages; beyond therapy, they also play a very important role in shaping microbial communities and evolution in the biosphere.

One of the key characteristics of phages is their host range, which is the set of bacterial hosts that a phage can infect. This host range is not fixed for a phage, it is determined by molecular interactions between the phage and the bacterial surface structures during the cycle of infection [1], [6]. Each phage attaches itself to the bacterial host, injects its viral genetic material which are able to replicate inside the host. Therefore, some of the phages have a small host range as they can only infect a limited number of bacterial strains while there are other which are capable of infecting a broad range of bacterias.

Identifying which phage infects which host, can have several applications in both environmental and clinical studies. In clinical, the host range helps us determine how suitable a phage is for designing the treatment for phage therapy. In environmental studies it is valuable to know the host being infected to understand the ecological interactions between viruses and microbial communities and evolution in the biosphere [6]. Precise host range knowledge is equally important for avoiding harm, an imprecise or overly broad host assignment risks eliminating bacteria that are not the intended target. This may include beneficial species, so correctly identifying a phage's host range is as much about protecting non target bacteria as it is about targeting the pathogens.

However, host range also brings its own obstacles. In phage therapy, we need to find the exact phage that infects the bacteria which is the cause of the infection, and for that we need a fast, reliable host prediction method. As noted in Chapter 1, traditional laboratory screening is accurate but expensive and time consuming, so realistically only a small portion of phage-host pairs can ever be tested, especially since many phages are difficult or impossible to culture in the laboratory. Metagenomic sequencing has bypassed this culturing barrier and revealed a vast number of new phages directly from the environment; as a result, the number of possible phage-host interactions has grown astronomically and computational predictions have become essential [2].

We formulate the host prediction problem as follows: let $\mathscr{X} \subseteq \mathbb{R}^d$ denote the feature space of phage representations, and let $\mathscr{H} = \{h_1, \ldots, h_N\}$ be the discrete label space of candidate bacterial hosts.

In a single-host classification setting, we observe random pairs $(X, H) \sim \mathbb{P}_{X,H}$, where $X \in \mathscr{X}$ is a feature vector representing a phage and $H \in \mathscr{H}$ denotes its true host. In a generalized multi-host setting, the target is a non-empty subset $H^* \subseteq \mathscr{H}$, i.e., $H^* \in \mathscr{P}(\mathscr{H}) \setminus \{\emptyset\}$, where $\mathscr{P}(\mathscr{H}) = 2^{\mathscr{H}}$ denotes the power set of $\mathscr{H}$.

Our primary objective is to construct a set valued prediction mapping

$$C : \mathscr{X} \longrightarrow \mathscr{P}(\mathscr{H}) \tag{2.1}$$

which maps an unlabelled test phage $X \in \mathscr{X}$ to a prediction set $C(X) \subseteq \mathscr{H}$ satisfying the finite sample coverage guarantee at a user specified significance level $\alpha \in (0,1)$:

$$\mathbb{P}(H \in C(X)) \geq 1 - \alpha. \tag{2.2}$$

where $\alpha \in (0,1)$ denotes the target error probability.

2.2 Current Computational Approaches and Limitations

The problem of predicting which phage infects which host bacteria has been studied computationally for over a decade and the available methods have significantly evolved during this time. These methods have progressed from simple rule based alignment to deep representation learning. Today, this field comprises of three main families of methods: the sequence similarity approaches, protein content based supervised classifiers, and protein language model (pLM) embedding approaches. There exist far more tools than we can review here [2], so we will only look at a few representative examples of each family, and conclude with the common limitation they share that motivates this work.

2.2.1 Sequence Similarity Methods

The early host prediction tools rely on sequence similarity between phage and bacterial genomes in two distinct ways, assuming that genomic similarity indicates infection compatibility. The first is **phage-host similarity**, $\mathrm{sim}(h, \mathrm{ph})$, we compare a phage genome directly against candidate host genomes, and then the host with the highest score is selected. The second is **phage-phage similarity**, $\mathrm{sim}(\mathrm{ph}_A, \mathrm{ph}_B)$, we compare a new phage against phages with already known hosts, and the host of the closest relative is transferred to it.

Phage–host similarity. WIsH [7] models each candidate host genome as a Markov chain with parameters θ_h and then selects the host that maximises the log likelihood of the new test phage sequence $S = (a_1, \ldots, a_L)$:

$$\hat{y} = \underset{h \in \mathscr{H}}{\operatorname{argmax}}\ \ell(S \mid \theta_h), \qquad \ell(S \mid \theta_h) = \sum_{i=k+1}^{L} \log \mathbb{P}(a_i \mid a_{i-k}, \ldots, a_{i-1};\, \theta_h).$$

Similarity is not explicitly computed here, the log likelihood $\ell(S \mid \theta_h)$ is used to measure how well the phage sequence fits the composition of each host genome.

Phage-phage similarity. HostPhinder [8] represents each genome g by its normalised k-mer frequency vector $v(g) \in \mathbb{R}^{4^k}$ and performs a nearest neighbour search over a reference database $\mathscr{D}_{\mathrm{ref}} = \{(g_i, y_i)\}_{i=1}^{N_{\mathrm{ref}}}$ of phages with known hosts:

$$\hat{y} = y\left(\arg\max_{g \in \mathscr{D}_{\mathrm{ref}}} \mathrm{sim}(v(g_{\mathrm{test}}), v(g))\right),$$

where $y(g)$ denotes the host label associated with reference genome $g \in \mathscr{D}_{\mathrm{ref}}$.

Both approaches offer practical advantages: they both do not require any protein level annotation, are computationally not expensive and can be applied to any newly sequenced phage or host genome. However, they share some notable limitations. Phage to phage transfer methods such as HostPhinder degrade sharply when a novel phage has no close relative in the reference database, since the entire prediction rests on a single nearest neighbour. Phage to host methods such as WIsH depend on the availability of well characterised host genomes. Neither of them exploits protein level functional information, which limits their ability to generalise beyond sequence composition patterns that they see in training. These two tools represent only a small sample of the methods developed in this space; there are works that exist beyond what is discussed here.

2.2.2 Protein Content Based Supervised Classifiers

To overcome the limitations of whole senome sequence alignment, protein content based supervised classifiers avoid sequence comparisons entirely by decomposing each phage into its constituent protein profile. They treat host prediction as a standard classification task. RaFAH [9] is an example of these type of methods, it works by clustering phage proteins into M protein clusters via MMseqs2 [10], then it represents each phage by a binary membership vector $\mathbf{x} \in \{0,1\}^M$, and trains a random forest to map this profile to a distribution over host genera:

$$f_{\mathrm{RF}} : \{0,1\}^M \longrightarrow \Delta^{|\mathscr{H}|-1}, \quad \text{where } \hat{p} = f_{\mathrm{RF}}(x), \quad \sum_{h \in \mathscr{H}} \hat{p}(h|x) = 1.$$

Unlike the sequence similarity methods of Section 2.2.1, RaFAH uses a trained classifier over a fixed protein cluster representation, which lets it generalise across a broader host range. But at the same time it brings the cost of requiring a protein clustering step and a labelled training set of phage host pairs, and inheriting whatever biases are present in that clustering.

2.2.3 Protein Language Models and Embedding-Based Methods

The current state of the art techniques use protein language models (pLMs), which are neural architectures trained on millions of protein sequences through self-supervised learning [11, 12]. Given a protein sequence $A = (a_1, \ldots, a_n) \in \mathscr{A}^n$ over the amino acid alphabet $\mathscr{A}$ of size 20, an encoder Φ_{pLM} produces a residue level embedding matrix $\mathbf{Z} = \Phi_{\mathrm{pLM}}(A) \in \mathbb{R}^{n \times d}$, which is then mean-pooled to a fixed-length sequence representation:

$$\mathbf{z} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{Z}_{i,:} \in \mathbb{R}^d.$$

Building on ProtT5 [11], EMPATHI [13] uses these embeddings to classify each individual phage protein into a functional category (e.g. capsid, tail fibre, lysin), and produces a per protein probability score for each candidate category. While SUBLYME

[14] applies a related embedding based approach to identify lysin proteins in metagenomic discovery. Neither of the tools predicts the bacterial host directly. Instead, these functional annotations can be used as the new raw material for our own pipeline. We aggregate the per protein functional probabilities produced by EMPATHI (Section 4.2) into a phage level, K-dimensional score vector $\mathbf{s}(X) = (s_1(X), \ldots, s_K(X))^T \in [0,1]^K$, where K is the total number of functional categories considered and each coordinate $s_k(X) \in [0,1]$ measures the aggregated evidence that phage X carries a protein of category k. The resulting vector then serves as the input to our conformal framework. This is consistent with earlier host prediction work using protein embeddings [15, 16].

2.2.4 Limitation: Absence of Uncertainty Quantification

Despite their empirical success, all of the methods seen above share a fundamental limitation: they produce point predictions (a single host label $\hat{y}$ or a scalar score $\hat{s}(x,y) \in [0,1]$) without any formal uncertainty quantification.

Uncertainty Quantification (UQ) is the mathematical framework used to rigorously measure, propagate, and bound prediction errors. A raw scalar output $\hat{s}(x,y) = 0.87$ merely reflects an uncalibrated relative ranking statistic, not a true conditional probability $\mathbb{P}(Y = y \mid X = x)$. Existing host prediction tools fail to provide UQ because no tool offers a finite sample statistical guarantee that the true bacterial host is contained within a candidate prediction set with a predefined target probability $1 - \alpha$.

Existing UQ frameworks only offer partial solutions. For example, Bayesian approaches require strong assumptions on the distributional form of the data that are difficult to justify for biological score data, while post-hoc calibration methods such as Platt scaling provide no finite-sample guarantees on unseen test data [4].

These limitations are consequential in phage therapy. In practice, a computational host prediction is not used to directly inject a phage into a patient, it is used to narrow a large candidate space down to a shortlist for laboratory confirmation before any clinical decision is made. However, an unreliable or overconfident prediction can still cause real harm at this trial stage. For example a false negative can exclude the true host from the shortlist entirely, which may delay the treatment while further candidates are screened, and a false positive can take up experimental validation time on a phage that was never going to work. This motivates the adoption of conformal prediction [3]: a distribution free framework that produces prediction sets which contain the true host with user specified probability $1 - \alpha$, and require only exchangeability of calibration and test data. Its formal construction follows in Section 2.3.

2.3 Conformal Prediction

Conformal prediction, initially developed by Vovk, Gammerman and Shafer [3] and then later made accessible by Angelopoulos and Bates [4], is a framework for constructing prediction sets that have a coverage guarantee. Unlike the methods seen in 2.2 it can provide us with a set of possible outcomes (not a point prediction), have a mathematical guarantee and yet requires no distributional assumptions on the data, except for the exchangibility condition. Here, we will formalize the framework of conformal prediction and analyze its finite-sample coverage properties.

Definition 2.3.1 (Exchangeability)**.** A finite sequence of random variables $Z_1, Z_2, \ldots, Z_m$ taking values in a space $\mathscr{Z}$ is exchangeable if, for every permutation σ of the indices $\{1, 2, \ldots, m\}$, the joint distribution of the permuted sequence is identical to that of the original sequence:

$$(Z_1, Z_2, \ldots, Z_m) \stackrel{d}{=} (Z_{\sigma(1)}, Z_{\sigma(2)}, \ldots, Z_{\sigma(m)})$$

where $\stackrel{d}{=}$ denotes equality in distribution.

Or, equivalently for all measurable sets $A_1, A_2, \ldots, A_m \subseteq \mathscr{Z}$ and all permutations $\sigma \in S_m$:

$$\mathbb{P}(Z_1 \in A_1, Z_2 \in A_2, \ldots, Z_m \in A_m) = \mathbb{P}(Z_{\sigma(1)} \in A_1, Z_{\sigma(2)} \in A_2, \ldots, Z_{\sigma(m)} \in A_m)$$

Independent and identically distributed (i.i.d.) sequences are always exchangeable, but the converse does not always hold. For example, if we draw two values without replacement from $\{0, 1\}$ we get the output $(Z_1, Z_2) \in \{(0, 1), (1, 0)\}$, both with probability $\frac{1}{2}$. This pair is exchangeable as its joint distribution does not change if we swap Z_1 and Z_2. But, the two variables are not independent, since conditioning on the first draw fully determines the second, $\left(\mathbb{P}(Z_2 = 1 \mid Z_1 = 1) = 0 \neq \mathbb{P}(Z_2 = 1) = \frac{1}{2}\right)$. Thus, exchangeability is a weaker condition than i.i.d., as it requires the observations to not have any special ordering. This condition can easily be met if we split any dataset randomly [5, 3].

2.3.1 Mathematical Setup and Prediction Set

Let $(X, H) \in \mathscr{X} \times \mathscr{H}$ be a random input-label pair, where $\mathscr{X}$ is the feature space that represents our object of interest (in our case it is the phage protein level score vectors) and $\mathscr{H} = \{h_1, \cdots, h_N\}$ is the discrete label space (set of candidate hosts).

We assume that we have a calibration dataset, $D_{\text{cal}} = \{(X_1, H_1), \cdots, (X_n, H_n)\}$, of n exchangeable data points drawn from an unknown joint distribution, $P_{X,H}$. Then, for a new unlabelled test input, X_{n+1}, our aim is to construct a prediction set function $C : \mathscr{X} \to 2^{\mathscr{H}}$ depending on the data which maps the test input to a subset of candidate labels, i.e., $C(X_{n+1}) \subseteq \mathscr{H}$, such that the true label falls in the set with a defined coverage that is:

$$\mathbb{P}(H_{n+1} \in C(X_{n+1})) \geq 1 - \alpha, \qquad \text{where } \alpha \in (0, 1) \text{ is the significance level}$$

Unlike point predictors, $C(X)$ dynamically reflects model uncertainty. It returns a singleton $|C(X)| = 1$ when confident, a multi-label set $|C(X)| \geq 2$ when uncertain between candidates, or an empty set $|C(X)| = 0$ for out-of-distribution inputs. The efficiency of the model can be quantified by the expected set size $\mathbb{E}[|C(X)|]$, an optimal predictor should achieve a coverage $\geq 1 - \alpha$ with the smallest possible set size.

2.3.2 Split conformal prediction

In split conformal prediction, the available data is divided into a training set and a disjoint calibration set, $D_{\text{cal}} = \{(X_i, H_i)\}_{i=1}^{n}$. The training set is used to fit an underlying scoring model $\hat{f} : \mathscr{X} \times \mathscr{H} \to \mathbb{R}$. We then define a nonconformity score function $s : \mathscr{X} \times \mathscr{H} \to \mathbb{R}$ based on $\hat{f}$. This score function quantifies how unusual a pair (x, h) is relative to the learned patterns, smaller values of $s(x, y)$ indicate stronger conformity between x and label h.

Then using the calibration set , $\{(X_i, H_i)\}_{i=1}^{n}$, we evaluate the nonconformity scores:

$$S_i = s(X_i, H_i), \quad \forall i = 1, \cdots, n.$$

To make sure that our error rate is less than α we need to compute the conformal calibration quantile, $\hat{q}$, from these calibration scores.

Definition 2.3.2 (Conformal Calibration Quantile)**.** The conformal calibration quantile q_α is defined as the empirical $\frac{\lceil (n+1)(1-\alpha) \rceil}{n}$-th quantile of calibration scores $\{S_1, \ldots, S_n\}$:

$$q_\alpha := \text{Quantile}\left(\{S_1, \ldots, S_n\} \, ; \, \frac{\lceil (n+1)(1-\alpha) \rceil}{n} \right)$$

Equivalently, expressed using the empirical cumulative distribution function:

$$q_\alpha = \inf \left\{ q \in \mathbb{R} : \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}(S_i \leq q) \geq \frac{\lceil (n+1)(1-\alpha) \rceil}{n} \right\} \; = \; S_{(\lceil (n+1)(1-\alpha) \rceil)}$$

where $S_{(1)} \leq S_{(2)} \leq \cdots \leq S_{(n)}$ denote the order statistics of the calibration scores, that is, the values $\{S_1, \ldots, S_n\}$ sorted in non decreasing order. With $k = \lceil (n+1)(1-\alpha) \rceil$, the counting function $\frac{1}{n} \sum_{i=1}^{n} \mathbf{1}(S_i \leq q)$ jumps from $\frac{k-1}{n}$ to $\frac{k}{n}$ exactly at $q = S_{(k)}$, so the smallest feasible q is the k-th smallest calibration score. In words, q_α is obtained by sorting the calibration scores and retaining the $\lceil (n+1)(1-\alpha) \rceil$-th smallest value.

Then, for a new input, X_{n+1} the conformal prediction set, $C(X_{n+1})$, is constructed by collecting all the candidate labels in $H \in \mathscr{H}$ whose nonconformity scores do not exceed q_α:

$$C(X_{n+1}) = \left\{ H \in \mathscr{H} \;\middle|\; s(X_{n+1}, H) \leq q_\alpha \right\} \tag{2.3}$$

The validity of this construction follows from exchangeability. By the definition 2.3.2, $q_\alpha = S_{(\lceil (n+1)(1-\alpha)\rceil)}$, where $S_{(1)} \leq \cdots \leq S_{(n)}$ are the order statistics of the calibration scores. Now let $S_{n+1} := s(X_{n+1}, H_{n+1})$ denote the nonconformity score of the true label of the new test point. Since the pairs $(X_1, H_1), \ldots, (X_{n+1}, H_{n+1})$ are exchangeable and the score function s is applied identically to every pair, the sequence of scores $(S_1, \ldots, S_n, S_{n+1})$ is exchangeable as well. All $(n+1)!$ relative orderings are equally likely, hence the rank of S within this sequence is uniformly distributed on $\{1, \ldots, n+1\}$. Consequently:

$$\mathbb{P}\big(S \leq q_\alpha\big) = \mathbb{P}(\text{rank}(S) \leq \lceil (n+1)(1-\alpha)\rceil) = \frac{\lceil (n+1)(1-\alpha)\rceil}{n+1} \geq 1-\alpha.$$

Theorem 2.3.1 (Marginal Coverage Guarantee [3, 17])**.** *If $(X_1, H_1), \ldots, (X_{n+1}, H_{n+1})$ are exchangeable, then the prediction set $C(X_{n+1})$ defined in Equation 2.3 satisfies:*

$$\mathbb{P}\left(H_{n+1} \in C(X_{n+1})\right) \geq 1-\alpha$$

Furthermore, if calibration scores $S_1, \ldots, S_n$ are almost surely distinct continuous random variables, the coverage bound is tight:

$$1-\alpha \leq \mathbb{P}\left(H_{n+1} \in C(X_{n+1})\right) \leq 1-\alpha + \frac{1}{n+1} \tag{2.4}$$

Crucially, theorem 2.3.1 holds strictly in finite samples without requiring parametric assumptions on $s(\cdot,\cdot)$ or $\mathbb{P}_{X,H}$. While the choice of scoring function s governs efficiency (set size), valid statistical coverage is guaranteed by construction in a single computational pass over D_{cal} [5].

The score function s is chosen so that it can separate plausible hosts from implausible ones. $s(x,h)$ should take a small value when h is likely to be the true host of x, and if h' is not the host, $s(x,h')$ should take a high score which typically exceeds q_α. h' then fails the test in Equation 2.3 and is not selected into the prediction set $C(X_{n+1})$. This discriminative behaviour is not created by conformal prediction itself, it is already exhibited by many standard supervised learning models. A classifier which is trained to output a class probability $\hat{p}(h \mid x)$ naturally produces this pattern since $1-\hat{p}(h \mid x)$ is small for the true class and large for incorrect ones. A poorly discriminating score function never invalidates the method as the coverage guarantee of Theorem 2.3.1 holds regardless, but the resulting prediction sets become larger.

2.3.3 Marginal & Conditional Coverage

Standard conformal prediction satisfies marginal coverage, which means that the $(1-\alpha)\%$ coverage guarantee holds on average across all possible choices of the calibration set, D_{cal}, and test samples, (X_{n+1}, H_{n+1}). So,

$$\mathbb{E}_{D_{cal},(X_{n+1},H_{n+1})}\left[\mathbb{I}(H_{n+1} \in C(X_{n+1}))\right] = \mathbb{P}(H_{n+1} \in C(X_{n+1})) \geq 1-\alpha$$

However, the marginal coverage does not ensure the conditional coverage, given specific features ($X = x$), or specific classes ($H = h$). So, for a dataset that has class imbalance, which is very common in phage-host interaction, a marginal model can satisfy the coverage guarantee by potentially over-covering a majority class (100%) and under-covering a small class (0%) [5, 18]. For instance, if $\mathscr{H} = \{-1, 1\}$ with class proportion $\mathbb{P}(H = -1) = 0.9$, a predictor achieving 100% coverage on $H = -1$ and 0% coverage on $Y = 1$ still attains an overall 90%. In phage-host classification, such conditional under-coverage on minority host may be a problem.

While feature-conditional coverage ($\mathbb{P}(H \in C(X) \mid X = x) \geq 1 - \alpha$) is mathematically impossible to achieve in finite samples without strong parametric assumptions [3, 5], class-conditional coverage (Mondrian conformal prediction) is exact and valid in finite samples. We partition D_{cal} into N disjoint subsets $D_{\text{cal}}^{(h)} = \{(X_i, H_i) \in D_{\text{cal}} : H_i = h\}$ and evaluate separate class-specific quantiles $q_\alpha^{(h)}$, enforcing exact coverage per class:

$$\mathbb{P}\left(H_{n+1} \in C(X_{n+1}) \,\middle|\, H_{n+1} = h\right) \geq 1 - \alpha, \quad \forall h \in \mathscr{H} \tag{2.5}$$

Because exchangeability holds within each class-conditional partition $D_{\text{cal}}^{(h)}$, evaluating per-class quantiles preserves exact finite-sample validity across every candidate host, as formalised in Section 3.5.2.

2.4 Conformal Set Aggregation (CSA)

The standard split conformal framework in 2.3.2 takes a scalar non conformity score $S(x, h) \in \mathbb{R}$. In this report, the protein language model pipeline (Section 2.2.3) produces a multidimensional representation, a K-dimensional score vector $\mathbf{s} \in [0, 1]^K$ for each phage, whose entries corresponding to specific functional categories (Figure 2.1). We may lose the geometric structure of the score distribution if we aggregate this vector to a scalar value before applying conformal prediction. For such settings, Ochoa Rivera, Patel and Tewari [5] have proposed a score-based aggregation (CSA) method for conformal prediction.

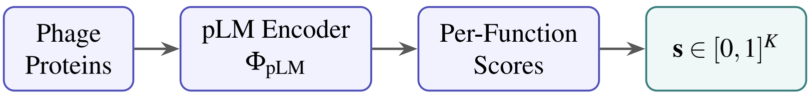

Figure 2.1: Construction of the K-dimensional functional score vector, $\mathbf{s}$, from a phage's protein sequences.

2.4.1 Ensemble Score Vectors

Let $\mathscr{S} \subseteq [0, 1]^K$ be the score space and let $s = (s_1, \cdots, s_K) \in \mathscr{S}$ be the vector of scores produced by the K models, for each phage. Now, the CSA method treats the entire vector as the object of inference rather than reducing it into a single value. The main idea is to define a region, called an envelope, $E \subseteq [0,1]^K$. This envelope consists of the region where the score vectors of the true hosts for that particular phage are concentrated.

Mathematically, let $\{\phi_h\}_{h\in\mathscr{H}}$ be a family of label-dependent maps $\phi_h : [0,1]^K \to [0,1]^K$ such that, when h is the true label, the transformed score vector $\phi_h(\mathbf{s})$ concentrates near the origin in $\mathbb{R}^K_+$. The CSA method constructs an envelope $E \subseteq \mathbb{R}^K_+$ as the intersection of M half-spaces induced by unit directions $\{\mathbf{u}_m\}_{m=1}^M$ and quantile thresholds $\{\tilde{q}_m\}_{m=1}^M$:

$$E = \bigcap_{m=1}^{M} \left\{\mathbf{s} \in \mathbb{R}^K_+ : \langle \mathbf{s}, \mathbf{u}_m \rangle \leq \tilde{q}_m \right\}. \tag{2.6}$$

By construction, E is convex. Membership of a transformed score in the scaled envelope $t \cdot E$ is tested coordinate-wise along each direction. The exact transformation ϕ_h used in this work is formalized in Section 3.3.

2.4.2 The CSA Framework

The core innovation of the CSA method is in the calibration process. It splits the calibration data into two subsets which have different roles. This is necessary for the exchangeability argument that provides us with the coverage guarantee. If we used the same calibration points to construct both, we would get overfitting which would introduce bias which would invalidate the distribution-free coverage guarantee [5].

(i) **Shape Discovery Set (S_1):** Used to model the spatial distribution of scores and learn the geometric shape, orientation, and boundaries of the envelope $E \subset \mathbb{R}^K$.

(ii) **Size Scaling Set (S_2):** Used to determine the scalar scaling parameter $\hat{t} \in \mathbb{R}^+$. $\hat{t}$ inflates or deflates E until exact empirical coverage $1-\alpha$ is attained over S_2:

$$\hat{t} = \inf\left\{ t > 0 \,\middle|\, \frac{1}{|S_2|} \sum_{\mathbf{s}\in S_2} \mathbf{1}\left(\phi_H(\mathbf{s}) \in t \cdot E\right) \geq \frac{\lceil (|S_2|+1)(1-\alpha) \rceil}{|S_2|} \right\}$$

This $\hat{t}$ is the smallest scaling factor such that the scaled envelope $\hat{t} \cdot E$ achieves the required $1-\alpha$ coverage. The scaled envelope is defined linearly as:

$$\hat{t} \cdot E = \{t \cdot x : x \in E\}, \quad \hat{t} > 0$$

In the original CSA paper [5], finding $\hat{t}$ is presented as a root-finding problem solved through iterative numerical optimization or binary search over candidate scaling values t. At test time, a candidate label h is included in the prediction set $C(\mathbf{s}_{n+1})$ if and only if its transformed score vector falls inside the scaled envelope:

$$C(\mathbf{s}_{n+1}) = \{h \in \mathscr{H} : \phi_h(\mathbf{s}_{n+1}) \in \hat{t} \cdot E\}.$$

2.4.3 Limitations of CSA in Classification

This CSA framework was originally developed and analyzed in a regression setting [5], and has some major limitations when it is applied to classification problems:

(i) **Convexity of Envelope:** In [5], the conformal envelope E is constructed as an intersection of half-spaces and is therefore convex. In the original high-dimensional regression setting, where the score vectors concentrate in a single connected region of the score space, this is a natural and useful design choice. But in classification, the score vectors for different classes concentrate in separate regions in the $[0,1]^K$ score space. Therefore, if we force a convex envelope across the regions, we will create a region that covers a large area of empty score space in between, which inflates the volume of the envelope and leads to over-conservative, uninformative prediction sets, as $|C(\mathbf{s})|$ becomes very large.

(ii) **Calibration Inefficiency:** Originally the method uses an iterative numerical optimization or a binary search to find the optimal scaling factor $\hat{t}$ over S_2, which is appropriate in the original regression analysis. Each trial value of t, however, requires evaluating every half-space constraint for every point in S_2, and in classification this cost multiplies across candidate host labels, making the iterative approach prohibitively expensive and motivating the closed-form computation of Section 3.5.

(iii) **Only Marginal Guarantee:** The scaling calculations in the original method only provide us with a marginal guarantee. This can lead to poor performance on the biological datasets, which are imbalanced.

(iv) **Handling of Missing Scores (NaN):** The original CSA formulation assumes that every score vector is fully observed across all K dimensions. In our setting, phage score vectors exhibit structural missingness (Section 4.1), a phage may simply lack a protein annotated for a given functional category. If we naively impute these missing entries, for example to zero, we would get a bias the nonconformity score toward interpreting absence as evidence against infection, and would invalidate the exchangeability assumption underlying Theorem 2.3.1.

These limitations directly motivate the methodology developed in Chapter 3.

— 3 —
Methodology

In this section, we will develop the mathematical framework used to construct prediction sets with guaranteed finite-sample phage-host classification. Section 3.1 formalises the prediction task and addresses missing data in our score vector. Then, building on the limitations of the CSA method identified in Section 2.4.3, Section 3.2 explains why non-convex envelopes are better suited for classification, and Section 3.3 defines the label-dependent nonconformity transformation used throughout this report to center valid score distributions near the origin. Section 3.4 then presents the three envelope geometries evaluated in this work, the two multi dimensional non-convex (Radial, Strip) and the collapsed 1D method. Finally, Section 3.5 derives the calibration scaling factor $\hat{t}$.

3.1 Problem Formalization

We will now apply the general conformal prediction setup from Section 2.3 to our host prediction task. Recall from Section 2.1 that $\mathscr{H} = \{h_1, \ldots, h_N\}$ represents the discrete set of N candidate bacterial hosts. Our goal is to build a function $C : \mathscr{X} \to \mathscr{P}(\mathscr{H})$ that outputs a set of hosts while maintaining the coverage guarantee in Equation (2.3).

In our work, the feature space, $\mathscr{X}$, does not use the raw phage sequences directly. Instead it consists of fixed-length score vectors (K dimensional) generated by the protein annotation process introduced in Section 2.2.3. So each score s_k corresponds to a specific functional protein category (e.g., capsid, tail, lysin), where K is the total number of categories. A specific phage genome may lack proteins in different functional categories, therefore the score vector may have structural missingness:

$$\mathbf{s} = (s_1, \ldots, s_K)^T \in ([0,1] \cup \{\mathrm{NA}\})^K.$$

Here, $s_k = \mathrm{NA}$ is the structural absence, which means that for category k no annotated protein was identified in the phage sequence, rather than a zero-evidence score or missingness caused by random measurement noise. This value is missing by design, not due to noise, and hence we need to handle it explicitly rather than simply imputing it.

To retain the mathematical guarantee without imputations, we attach a missingness mask, $\mathbf{m}$, to every score vector. We represent each score observation as a paired vector $(\mathbf{s}, \mathbf{m}) \in ([0,1] \cup \{\mathrm{NA}\})^K \times \{0,1\}^K$:

$$\mathbf{m} = (m_1, \ldots, m_K)^T \in \{0,1\}^K, \qquad m_k = \mathbf{1}(s_k \neq \mathrm{NA}).$$

Given a calibration set $D_{\mathrm{cal}} = \{(\mathbf{s}_i, \mathbf{m}_i, H_i)\}_{i=1}^n$ of exchangeable samples, and an unlabelled test sample $(\mathbf{s}_{n+1}, \mathbf{m}_{n+1})$ drawn exchangeably from an unknown joint probability distribution, our goal is to construct a set-valued prediction mapping $C : ([0,1] \cup \{\mathrm{NA}\})^K \times \{0,1\}^K \to \mathscr{P}(\mathscr{H})$ at significance level $\alpha \in (0,1)$ such that:

$$\mathbb{P}\Big(H_{n+1} \in C(\mathbf{s}_{n+1}, \mathbf{m}_{n+1})\Big) \geq 1 - \alpha.$$

The total number of observed functional dimensions for a given phage is given by the L_1 norm of the mask, $\|\mathbf{m}\|_1 = \sum_{k=1}^K m_k$. Then we have $\mathbf{s}_{\mathbf{m}} \in [0,1]^{\|\mathbf{m}\|_1}$, the sub-vector restricted strictly to the observed dimensions where $m_k = 1$. All prediction functions and nonconformity scoring operators developed in this report are defined over the product space of pairs $(\mathbf{s}, \mathbf{m})$, rather than $\mathbf{s}$ alone. This ensures our nonconformity scores and envelope tests remain mathematically valid for any missingness pattern, not just fully observed data, $\mathbf{m} = \mathbf{1}$.

We apply this framework to two host prediction tasks:

(i) **Gram-type classification** ($N = 2$): $\mathscr{H} = \{\text{gram-neg}, \text{gram-pos}\}$, evaluated over K functional category dimensions.

(ii) **Host-level classification** ($N > 2$): $\mathscr{H}$ represents the full set of bacterial hosts in our dataset. For each host $h \in \mathscr{H}$, the score space is decomposed into host-specific sub-vectors of dimension K_h, and an individual envelope E_h is fitted independently on training phages with true host h.

In practice, host-level classification is not evaluated over the full host space $\mathscr{H}$ at test time. We first obtain a gram-type prediction set $C_{\mathrm{gram}}(s,m) \subseteq \{\text{gram-neg}, \text{gram-pos}\}$ Section 3.3.1, and use it to restrict the candidate host set to

$$\mathscr{H}_{\mathrm{cand}}(s,m) = \bigcup_{g \in C_{\mathrm{gram}}(s,m)} \mathscr{H}_g \subseteq \mathscr{H},$$

where $\mathscr{H}_g \subset \mathscr{H}$ denotes the subset of hosts of the true gram type g. Let $(s_h, m_h) \in ([0,1] \cup \{\mathrm{NA}\})^{K_h} \times \{0,1\}^{K_h}$ denote the restriction of s to the K_h score dimensions associated with host h.

Once we construct the per host envelopes E_h and the calibration thresholds $\hat{t}_h$ (detailed in Section 3.4 and 3.5), the host level prediction set is given by:

$$C_{\text{host}}(s,m) = \{h \in \mathscr{H}_{\text{cand}}(s,m) \mid \phi_h(s_h,m_h) \in \hat{t}_h \cdot E_h\}\,. \tag{3.1}$$

rather than a single-stage test over all of $\mathscr{H}$. This cascade reduces the number of per-host envelope evaluations required at test time, at the cost of making C_{host} dependent on the upstream gram-type prediction.

The following result confirms that composing the gram type and host level stages does not degrade the coverage guarantee of Equation (3.1).

Proposition 3.1.1 (Cascade Validity). *Under exchangeability, if gram type classification is class conditionally valid at level α and each candidate host envelope satisfies validity level α_h, the cascaded predictor satisfies*

$$\mathbb{P}\big(H^\star \notin C_{host}(\mathbf{s})\big) \leq \alpha + \max_{h\in\mathscr{H}} \alpha_h. \tag{3.2}$$

Proof. The miscoverage event $\{h^\star \notin C_{\text{host}}(\mathbf{s})\}$ implies either that the true host was pruned by the Gram type filter ($h^\star \notin \mathscr{H}_{\text{cand}}(\mathbf{s})$, which requires $g^\star \notin C_{\text{gram}}(\mathbf{s})$) or that $h^\star$ was admitted to the candidate pool but rejected by its host envelope ($\tau_{h^\star}(\mathbf{s}) > \hat{t}_{h^\star}$). Applying the union bound:

$$\begin{aligned}
\mathbb{P}\big(h^\star \notin C_{\text{host}}(\mathbf{s})\big) &\leq \mathbb{P}\big(g^\star \notin C_{\text{gram}}(\mathbf{s})\big) + \mathbb{P}\big(\tau_{h^\star}(\mathbf{s}) > \hat{t}_{h^\star}\big) \\
&= \sum_g \mathbb{P}(G=g)\,\mathbb{P}(g^\star \notin C_{\text{gram}} \mid G=g) + \sum_h \mathbb{P}(H=h)\,\mathbb{P}(\tau_h > \hat{t}_h \mid H=h) \\
&\leq \alpha + \max_{h\in\mathscr{H}} \alpha_h.
\end{aligned}$$

□

3.2 Non-Convex Envelopes in Classification

Section 2.4.3 identified envelope convexity as one of the main limitations of standard CSA method [5] when applied to classification. While convex envelopes naturally model continuous target spaces in regression, classification score vectors concentrate in disjoint, class-specific clusters across the score space. Therefore, it leads to the envelope being stretched over regions of empty spaces, making the prediction sets unnecessarily large. Here, we will describe the alternate method that we have used in this report.

In classification, score vectors for different classes occupy separated regions in the score space. If we attempt to fit a single shared convex envelope, we would force the boundary to bridge across empty intermediate space, which would lead to uninformatively large prediction sets. To prevent this, we fit dedicated class conditional envelopes E_h for each candidate hos. This allows the boundary to closely adapt to the non linear structure of true infector scores without the convex over extension.

Also very importantly, dropping convexity does not violate statistical validity: the conformal coverage guarantee holds for any measurable envelope $E \subseteq \mathbb{R}_+^K$ under exchangeability [5]. To ensure that the boundary is both computationally tractable and biologically sensible, the envelope must satisfy two geometric properties:

1. **Star shapedness with respect to the origin:** To ensure the scaling function $\tau(\mathbf{s}) = \inf\{t > 0 : \mathbf{s} \in t \cdot E\}$ is continuous and yields a unique scalar threshold.
2. **Coordinate wise monotonicity (convexity toward the axes):** In nonconformity space, the origin $\mathbf{0}$ represents perfect conformity (maximum infection evidence). If a score vector $\mathbf{x} \in E$ is accepted, any point $\mathbf{y}$ with $0 \leq y_k \leq x_k$ for all k exhibits even stronger evidence and must also be accepted.

To ensure that the boundary is both computationally tractable and biologically sensible, the envelope must satisfy two geometric properties: star shapedness (Definiton 3.2.1) with respect to the origin (ensuring a unique, continuous scaling threshold $\tau(s)$) and coordinate wise monotonicity (Definition 3.2.2) which ensures convexity toward the axes, so stronger conformity vectors closer to the origin are unconditionally included.

Definition 3.2.1 (Star-Shaped Set)**.** A set $E \subseteq \mathbb{R}^K$ is said to be *star-shaped with respect to the origin* $0 \in \mathbb{R}^K$ if for every point $x \in E$ and every scalar $\lambda \in [0,1]$, the line segment connecting the origin to x lies entirely within E:

$$\lambda x \in E \quad \forall x \in E,\ \forall \lambda \in [0,1].$$

Definition 3.2.2 (Coordinate Wise Monotone Set)**.** A set $E \subseteq \mathbb{R}_+^K$ is coordinate wise monotone if for all $\mathbf{x} \in E$ and $\mathbf{y} \in \mathbb{R}_+^K$:

$$\mathbf{0} \leq \mathbf{y} \leq \mathbf{x} \implies \mathbf{y} \in E,$$

where $\mathbf{y} \leq \mathbf{x}$ denotes $y_k \leq x_k$ for all coordinates $k \in \{1, \dots, K\}$.

3.3 Nonconformity Score Transformation

The raw score vectors are not directly comparable across candidate labels and cannot be used directly as nonconformity measures. Recall from Section 2.4.1 that the CSA framework requires a label-dependent transformation $\phi_h : ([0,1] \cup \{\texttt{NA}\})^K \times \{0,1\}^K \to (\mathbb{R}_+ \cup \{\texttt{NA}\})^K$ that maps candidate hypotheses into a nonconformity score space centered near the origin.

The raw functional category scores $s_{k,h} \in [0,1]$ quantify evidence in favor of functional compatibility, therefore a raw score close to 1 reflects strong infection compatibility (low nonconformity), whereas a score near 0 reflects high nonconformity. Thus, the mapping is defined such that when a candidate hypothesis $h \in \mathscr{H}$ corresponds to the true host label, the resulting vector $\mathbf{v} = \phi_h(\mathbf{s}, \mathbf{m})$ concentrates near $\mathbf{0} \in \mathbb{R}_+^K$, whereas incorrect hypotheses yield vectors situated significantly further from the origin. We now formalize ϕ_h for both classification settings while explicitly preserving structural missingness via the mask vector $\mathbf{m}$.

To map all the score vectors into a nonconformity space near the origin, we define class dependent transformations ϕ_h. Formally, for a single score dimension $s_k \in [0,1]$ with the binary mask $m_k \in \{0,1\}$, we define the general coordinate wise mapping:

$$\phi_h(s_k, m_k) = \begin{cases} |s_k - c_h| & \text{if } m_k = 1, \\ \mathtt{NA} & \text{if } m_k = 0, \end{cases} \qquad c_h \in \{0,1\}, \tag{3.3}$$

where every hypothesis uses $c_h = 1$ to measure deviation from maximal evidence. For any observed dimension ($m_k = 1$), this simplifies directly to $\phi_h(s_k) = 1 - s_{h,k}$, this ensures that $\phi_h(s_k) \to 0$ whenever $s_{h,k} \to 1$.

3.3.1 Gram Type Classification Nonconformity Mapping

In the Gram type classification setting ($N = 2$, $\mathscr{H} = \{\text{gram-neg}, \text{gram-pos}\}$), we treat the task as a standard two class classification problem where each candidate hypothesis $h \in \mathscr{H}$ is evaluated on a candidate score vector $s_h \in ([0,1] \cup \{\text{NA}\})^K$. The functional models output a Gram positive evidence score $s \in [0,1]^K$, therefore the we write the class specific evidence vectors as $s_{\text{grampos}} = s$ and $s_{\text{gramneg}} = 1 - s$. A high score represents strong evidence in favor of hypothesis h, so we have $c_h = 1$ for all classes. After applying $\phi_h(s_h, m_h)_k = 1 - s_{h,k}$ we get the host specific mapping across $k \in \{1, \ldots, K\}$:

$$v_k = (\phi_h(\mathbf{s}, \mathbf{m}))_k = \begin{cases} s_k & \text{if } h = \text{gram-neg and } m_k = 1, \\ 1 - s_k & \text{if } h = \text{gram-pos and } m_k = 1, \\ \mathtt{NA} & \text{if } m_k = 0. \end{cases} \tag{3.4}$$

Using (3.4), we get a high nonconformity score if we test an incorrect hypothesis, and the scores for the true hypothesis are mapped near the origin $\mathbf{0} \in \mathbb{R}_+^K$. This allows both classes to be calibrated against a shared reference envelope $E \subset \mathbb{R}_+^K$.

3.3.2 Host Level Classification Nonconformity Mapping

In the host level setting ($N > 2$), the candidate labels do not share the scores. Instead, each candidate host $h \in \mathscr{H}$ is evaluated on a host-specific sub-vector $\mathbf{s}_h \in ([0,1] \cup \{\text{NA}\})^{K_h}$. This measures the direct functional compatibility with host h.

A high raw score $s_{h,k}$ indicates high likelihood of infection, $c_h = 1$ for all hosts, using this we can get the following host-specific mapping across $k \in \{1, \dots, K_h\}$:

$$v_{h,k} = (\phi_h(\mathbf{s}_h, \mathbf{m}_h))_k = \begin{cases} 1 - s_{h,k} & \text{if } m_{h,k} = 1, \\ \text{NA} & \text{if } m_{h,k} = 0. \end{cases} \tag{3.5}$$

Unlike the Gram classification task, in host level prediction we construct N separate host envelopes $E_h \subset \mathbb{R}_+^{K_h}$ which are fitted independently on calibration phages with true label $H = h$. At test time, $\mathbf{s}_h$ is transformed using (3.5) and tested against E_h, this avoids forcing unrelated host signatures into a single shared coordinate system.

Both transforms are strictly monotone fixed maps that require no parameter estimation from data. Therefore, they do not have sample dependencies that could break the exchangeability argument (which is necessary for the coverage guarantee, Theorem 2.3.1). The missingness masks $\mathbf{m}$ propagate unchanged as `NA` values, and the envelope constructions of Section 3.4 operate directly on this representation.

3.4 Envelope Construction

Here we will present the three envelope geometries that we have evaluated in this report. Each method follows the two-stage calibration procedure: the shape-discovery set S_1 determines the geometric form of the envelope, and the size scaling set S_2 determines the scaling factor $\hat{t}$. All three envelopes are star-shaped with respect to the origin, so the per-point scaling function

$$\tau(\mathbf{s}) = \inf\{t > 0 : \mathbf{s} \in t \cdot E\} \tag{3.6}$$

is well-defined as a single scalar for every $\mathbf{s} \in \mathbb{R}_+^K$. For each envelope type, we derive a closed form expression for $\tau(\mathbf{s})$ that eliminates the need for iterative search. The scaling factor $\hat{t}$ is then computed using class conditional quantile aggregation over S_2, as detailed in Section 3.5.

3.4.1 Collapsed 1D Envelope Method

The simplest envelope construction method just collapses the K-dimensional (or K_h-dimensional) NC vector to a scalar value by taking the mean over the over observed dimensions,

$$\bar{s} = \frac{1}{|\mathbf{m}|_1} \sum_{k:m_k=1} v_k,$$

and calibrates a single quantile on S_1:

$$\tilde{q} = \text{Quantile}_{1-\alpha}\left(\{\bar{s}_i\}_{i \in S_1}\right), \qquad \tau(\mathbf{s}) = \frac{\bar{s}}{\tilde{q}}, \qquad E = \{\mathbf{s} : \bar{s} \leq \tilde{q}\}.$$

This envelope is star-shaped (Section 3.2), it is unaffected by the dimensionality K, but at the same time is also discards all the geometric information that the K dimensions provide us with. Any two vectors with the same scalar average are treated identically regardless of their coordinate wise distribution. Consequently, if a single functional dimension has a highly confident infection signal (e.g., $s_{h,k} \approx 1$, so $v_k \approx 0$) while the others contains a small one, mean pooling dilutes this primary signal across the remaining axes. In contrast, the multidimensional non convex envelopes developed in Sections 3.4.2 and 3.4.3 preserve the coordinate wise geometry, they are able to isolate the strong single dimension evidence from noisy background dimensions.

3.4.2 Radial Envelope Method

The Radial envelope method retains the directional structure of the scores by calibrating a boundary radius as a function of direction on the unit sphere which is restricted only to the positive orthant (since all NC values are non-negative). In this method, we draw M directional vectors $\{\mathbf{u}_m\}_{m=1}^{M}$ uniformly over the orthant by sampling $\mathbf{v}_m \sim |\mathcal{N}(0, I_K)|$ coordinate-wise and normalizing: $\mathbf{u}_m = \mathbf{v}_m / \|\mathbf{v}_m\|$.

For each S_1 point $\mathbf{s}_i$, over the observed dimensions only, we compute its magnitude and unit direction, $r_i = \|\mathbf{s}_{i,\mathbf{m}_i}\|$, $\mathbf{d}_i = \mathbf{s}_{i,\mathbf{m}} / r_i$. For each sample direction $\mathbf{u}_m$, we collect n_m calibration points within an angular sector of half-width $\delta = 30°$ (where $\mathbf{d}_i \cdot \mathbf{u}_m \geq \cos\delta$). The directional radius $\tilde{q}_m$ is then calibrated as:

$$\tilde{q}_m = \begin{cases} \text{Quantile}_{1-\alpha}\left(\{r_i : \mathbf{d}_i \cdot \mathbf{u}_m \geq \cos\delta\}\right) & \text{if } n_m \geq n_{\min}, \\ w_m \cdot \text{Quantile}_{1-\alpha}\left(\{r_i : \mathbf{d}_i \cdot \mathbf{u}_m \geq \cos\delta\}\right) + (1 - w_m) \cdot \tilde{q}_{\text{marg}} & \text{if } 0 < n_m < n_{\min}, \\ \tilde{q}_{\text{marg}} & \text{if } n_m = 0, \end{cases}$$

where n_m is the number of S_1 points within the band of direction m, $n_{\min}$ a minimum-support threshold, $w_m = n_m / n_{\min}$, and $\tilde{q}_{\text{marg}} = \text{Quantile}_{1-\alpha}(\{r_i\}_{i \in S_1})$ the marginal fallback radius. This blend enables us to avoids a hard cutoff between sufficient and insufficient local support (number of points in that region).

To ensure that we get a smooth boundary, we compute the radius at an arbitrary direction $\mathbf{d}$ as a directional weighted average of all M calibrated radii:

$$r(\mathbf{d}) = \sum_{m=1}^{M} w_m(\mathbf{d})\, \tilde{q}_m, \qquad w_m(\mathbf{d}) = \frac{\exp(\kappa\, \mathbf{d} \cdot \mathbf{u}_m)}{\sum_{m'=1}^{M} \exp(\kappa\, \mathbf{d} \cdot \mathbf{u}_{m'})}, \tag{3.7}$$

where $\kappa > 0$ controls the angular smoothness of the interpolation across directions (larger κ approaches nearest-direction assignment). This makes the envelope boundary a continuous function of direction rather than a piecewise constant one. The scaling function follows directly from this:

$$\tau(\mathbf{s}) = \frac{\|\mathbf{s_m}\|}{r(\mathbf{d})}, \qquad \mathbf{d} = \frac{\mathbf{s_m}}{\|\mathbf{s_m}\|}, \qquad E = \{\mathbf{s} : \|\mathbf{s_m}\| \leq r(\mathbf{d})\}.$$

As established in Section 3.2, this construction is star-shaped by design. This is because r depends only on direction and not on magnitude, $\mathbf{s} \in t \cdot E$ for some t implies $\lambda \mathbf{s} \in t \cdot E$ for all $\lambda \in [0,1]$. Figure 3.1 illustrates this construction, sample directions u_m collect calibration points within angular cones of half angle δ, establishing local radii $\tilde{q}_m$ that are kernel smoothed into a continuous boundary.

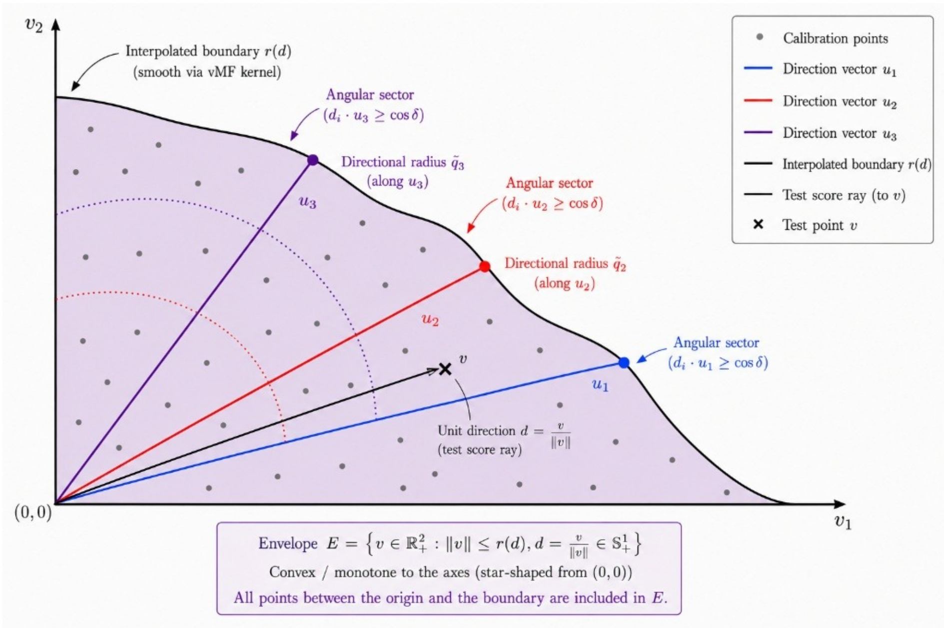

Figure 3.1: Geometric construction of the radial envelope in 2D score space. The score space is partitioned into angular sectors around reference rays $\theta_1, \theta_2, \theta_3$. For each sector, the empirical $(1-\alpha)$ quantile radius $\hat{q}_{1-\alpha}(\theta)$ is computed from calibration scores (gray dots) to define the envelope boundary $L(\theta)$ (solid black curve). A test score vector v with polar coordinates (r, θ_v) is included in the conformal region if its radial magnitude satisfies $r = \|v\| \leq L(\theta_v)$.

3.4.3 Strip Envelope Method

The Strip envelope defines conditional bounds for each target dimension i based on the bin occupied by another dimension j. We partition the range of dimension j into NB equal-width bins. For each bin n containing $n_{j,n}$ calibration points, the upper limit along dimension $i \neq j$ is calibrated as:

$$\text{limits}[j,i,n] = \begin{cases} \text{Quantile}_{1-\alpha}\left(\{s_{\cdot,i} : s_{\cdot,j} \in \text{bin } n\}\right) & \text{if } n_{j,n} \geq n_{\min}, \\ w_{j,n} \cdot \text{Quantile}_{1-\alpha}\left(\{s_{\cdot,i} : s_{\cdot,j} \in \text{bin } n\}\right) + (1-w_{j,n}) \cdot \tilde{q}_i & \text{if } 0 < n_{j,n} < n_{\min}, \\ \tilde{q}_i & \text{if } n_{j,n} = 0, \end{cases}$$

where $\tilde{q}_i$ is the marginal $(1-\alpha)$ quantile for dimension i over S_1 , and $w_{j,n} = n_{j,n}/n_{\min}$ shrinks underpopulated bins toward it (it mirrors the blending rule of the radial envelope).

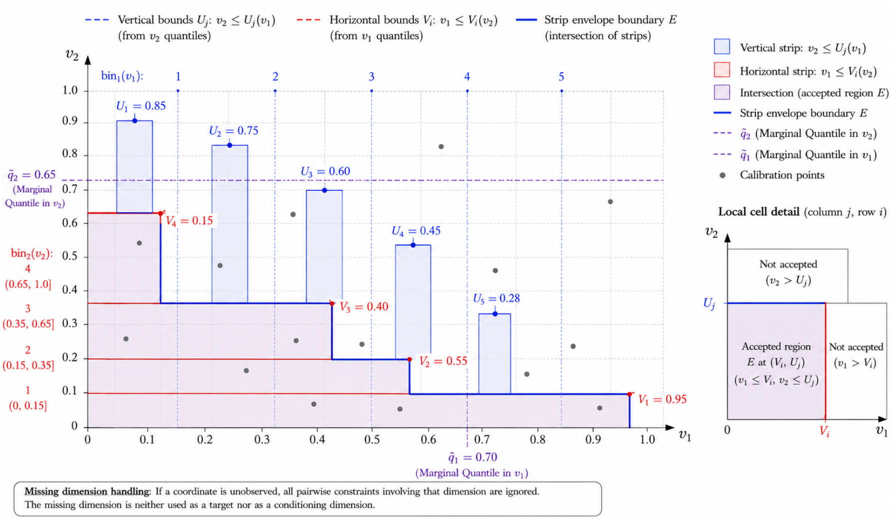

Figure 3.2: Strip envelope 2D staircase construction and missing dimension handling. Dimension v_1 is partitioned into equal width bins with vertical upper bounds U_j, while dimension v_2 defines horizontal bounds V_i. The intersection of both directional constraints forms the coordinate wise monotone staircase envelope E (shaded region). If a coordinate is unobserved, all pairwise constraints involving that dimension are ignored, so the missing dimension is neither used as a target nor as a conditioning dimension.

A windowed monotonicity pass then enforces $\text{limits}[j,i,n] \geq \text{limits}[j,i,n']$ for $n' \in (n, n+w]$ within a fixed window w.

$$\text{limits}[j,i,n] \leftarrow \max_{n' \in [n, \min(n+w, NB-1)]} \text{limits}_{\text{raw}}[j,i,n'].$$

This is done to ensure that a bin's limit cannot be undercut by a bin immediately ahead of it in the same conditioning dimension. The monotonicity is necessary to ensure that a phage with a smaller value (more confident) in the conditioning dimension j, does not recieve a tighter bound in dimension i than a phage with a larger value in j, which makes sure that the conditional envelope is logically correct. This ensures that a single sparse or fallback-driven bin at one end of the range cannot set the limit for every bin behind it. This report uses $w \geq NB$ for gram type classification and $w = 2$ for host-level classification, reflecting the difference in how many bins (NB) each task typically calibrates.

Geometrically, the strip envelope forms a coordinate wise monotone staircase region defined by the intersection of conditional bin regions across all dimension pairs:

$$E = \bigcap_{j=1}^{K} \bigcap_{i \neq j} \bigcup_{k=1}^{N_B} \left\{ v \in \mathbb{R}_{+}^{K} : v_i \leq \text{limits}[j,i,k] \right\}. \tag{3.8}$$

For a score vector v with an observed dimension set $\mathscr{A}(v)$, the Strip scaling function is

$$\tau(v) = \max_{\substack{j,i \in \mathscr{A}(v) \\ j \neq i}} \frac{v_i}{\text{limits}[j,i,\text{bin}_j(v_j)]}.$$

So, a pairwise constraint contributes to $\tau(v)$ if and only if both its target and conditioning dimensions are observed:

$$(j,i) \text{ contributes to } \tau(v) \quad \Longleftrightarrow \quad m_j = m_i = 1, \quad j \neq i.$$

If either dimension is missing,

$$m_j = 0 \quad \text{or} \quad m_i = 0,$$

the corresponding pairwise constraint is omitted from the maximum.

Structural missingness is handled by dropping dimensions that are unobserved for the score vector under consideration. Let

$$\mathscr{A}(v) = \{ i \in \{1, \ldots, K\} : m_i = 1 \}$$

be the set of observed dimensions of a score vector v, where $m_i \in \{0,1\}$ is its observation mask.

For a missing dimension $i \notin \mathscr{A}(v)$, we discard all the conditional limits for which i is the conditioning dimension:

$$\text{limits}[i,j,k] \text{ is ignored for all } j \neq i, \; k = 1, \ldots, N_B.$$

Similarly, the limits for which i is the target dimension are ignored because the target value v_i is unobserved:

$$\text{limits}[j,i,k] \text{ does not contribute to } \tau(v), \qquad i \notin \mathscr{A}(v).$$

Therefore, the Strip construction is evaluated only over pairs of observed dimensions:

$$j,i \in \mathscr{A}(v), \qquad j \neq i.$$

3.5 $\hat{t}$ Calculation

3.5.1 τ Computation

For any test point $\mathbf{s} \in \mathbb{R}^K$, we define the inclusion threshold function $\tau(\mathbf{s})$ (defined in 3.6) as the minimum scaling factor required for the scaled envelope $t \cdot E$ to contain $\mathbf{s}$:

$$\tau(\mathbf{s}) = \inf\{t > 0 : \mathbf{s} \in t \cdot E\}$$

Given the computed τ values over the scaling set S_2, the calibrated scaling factor $\hat{t}$ is selected as the $(1-\alpha)$ empirical quantile:

$$\hat{t} = \tau_{(k)} \quad \text{where} \quad k = \lceil (|S_2|+1)(1-\alpha) \rceil$$

Here, $\tau_{(k)}$ denotes the k-th order statistic (the k-th smallest value) of $\{\tau(\mathbf{s}_i)\}_{\mathbf{s}_i \in S_2}$.

Unlike the original CSA method, which relies on a computationally expensive iterative binary search over candidate t values, our closed form expressions allow $\tau(\mathbf{s})$ to be evaluated directly for each point in a single pass over S_2. This eliminates the iterative search entirely, which reduces the calibration computational complexity from $O(|S_2| \cdot N_{\text{iter}})$ to $O(|S_2| \log |S_2|)$.

3.5.2 Class Specific $\hat{t}$ Calculation

When the dataset has class imbalance (like our dataset), a single global scaling factor $\hat{t}$ can cause under coverage for complex classes and over coverage for simpler ones. To enforce exact class conditional coverage $1-\alpha$, we partition S_2 into class-specific calibration sets $S_2^{(c)} = \{\mathbf{s} \in S_2 : h = c\}$ for each label $c \in \mathscr{H}$.

For each class c, the class-specific scaling threshold $\hat{t}_c$ is calculated independently on $S_2^{(c)}$:

$$\hat{t}_c = \tau^{(c)}_{(k_c)}, \qquad k_c = \lceil (|S_2^{(c)}|+1)(1-\alpha) \rceil. \tag{3.9}$$

The two prediction tasks in this report differ in how envelopes are constructed, which determines how these per class thresholds $\hat{t}_c$ are applied:

Gram Type Classification: We only fit a single envelope, E on the shared shape discovery set S_1 containing phages of both classes. If we evaluate a single pooled threshold over S_2, it would only yield only marginal coverage. So, we compute the class specific quantiles $\hat{t}_{\text{gram-neg}}$ and $\hat{t}_{\text{gram-pos}}$ using (3.9) and scale the shared envelope by their maximum:

$$\hat{t} = \max\left(\hat{t}_{\text{gram-neg}}, \hat{t}_{\text{gram-pos}}\right). \tag{3.10}$$

Since $\hat{t} \geq \hat{t}_c$ for both classes, scaling E by $\hat{t}$ satisfies both coverage requirements simultaneously, guaranteeing class-conditional validity $\mathbb{P}(\mathbf{s} \in \hat{t} \cdot E \mid H = c) \geq 1 - \alpha$ for $c \in \{\text{gram-neg}, \text{gram-pos}\}$.

Host Level Classification: In host-level classification, each candidate host $h \in \mathscr{H}$ has a dedicated envelope E_h fitted exclusively on phages with true host h (Section 3.1). Because its calibration set $S_2^{(h)}$ is already single-class, $\hat{t}_h$ is computed directly via 3.9 without taking a maximum across hosts. Each scaled envelope $\hat{t}_h \cdot E_h$ is validated independently, admitting a candidate host h if $\phi_h(\mathbf{s}_h) \in \hat{t}_h \cdot E_h$.

3.5.3 Resolving the Empty Prediction Set

In section 3.1 we noted that $C(\mathbf{s}, \mathbf{m})$ may be empty when no candidate label's transformed score falls inside $\hat{t} \cdot E$. In both our tasks we resolve this differently.

For gram type classification, we resolve an empty set conservatively by returning both labels, $C(\mathbf{s}) = \mathscr{H}$, rather than an arbitrary tie break, since $N = 2$, this is the uninformative but always valid choice. On the other hand, for host-level classification, where N can be large and returning all of $\hat{\mathscr{H}}(\mathbf{s})$ would be uninformative, we resolve an empty set by forcing a singleton at the host with the closest τ value:

$$C(\mathbf{s}) = \left\{ \underset{h \in \hat{\mathscr{H}}(\mathbf{s})}{\operatorname{argmin}} \, \tau_h(\mathbf{s}_h) \right\}.$$

Both apply only when the primary rule of Equation (2.3) yields $\emptyset$. Coverage guarantee in Theorem 2.3.1 remains unaffected as nonempty fallback options strictly enlarge $C(\mathbf{s})$, thus the lower bound $\mathbb{P}(H \in C(\mathbf{s})) \geq 1 - \alpha$ trivially holds. Consequently, the reported `empty_rate` is strictly zero by rule rather than an absence of out-of-envelope points. To preserve visibility of these points, Section 5 reports the pre-resolution `forced_rate` separately.

— 4 —

Data and Preprocessing Pipeline

In this section, we describe the dataset architecture 4.1, the mathematical aggregation operators used to combine protein-level functional annotations into phage-level score vectors 4.2, and the masking strategy developed to eliminate bias caused by cross-fold protein cluster contamination 4.3.

4.1 Data Description and Structural Missingness

Our dataset comprises of $N_{\text{prot}} = 1\,853\,074$ protein sequences extracted from 18 474 unique bacteriophages. The feature extraction and score generation pipeline is structured in five stages (Figure 4.1):

1. **Genome Translation:** Each phage genome sequence is decomposed into its individual translated protein sequences.
2. **Sequence Embeddings:** Protein sequences are encoded into numerical representations using protein language models.
3. **Hierarchical Functional Annotation (EMPATHI):** Using the EMPATHI architecture [13], protein embeddings are classified to predict broad categories first (e.g., structural, replication, lysis) to constrain specific sub functions (e.g., tail fiber, capsid, lysin, DNA polymerase).
4. **Host Function Base Classifiers ($M_{h,k}$):** For each candidate host $h \in \mathscr{H}$ $(N = 54)$ and functional category k ($K_h \leq 40$), an independent supervised classifier $M_{h,k}$ is trained on the functional embeddings. Each classifier outputs a probability score $s_{p,h,k} \in [0,1]$ measuring the compatibility of protein p with host h for functional role k, yielding $M = 2{,}160$ base models. Not every host has valid models for all 40 categories, so the per host dimensionality is not fixed and K_h ranges from 26 to 40. Summing K_h over all 54 hosts gives the final dimensionality, $K_{\text{tot}} = 2{,}036$, which is smaller than $M = 2{,}160 = 54 \times 40/$

5. **Phage-Level Aggregation:** For each phage genome, predictions across all its constituent proteins are aggregated via dimension wise, NaN aware mean pooling (Section 4.2), producing a $K_{\text{tot}} = 2{,}036$-dimensional phage score vector.

In this work, these pre computed scores serve as fixed inputs to our conformal framework without retraining the underlying classifiers. Each protein sequence is linked to its phage accession, host genus label $H \in \mathscr{H}$ ($N = 54$ genera) and host Gram type $G \in \{\text{gram-neg}, \text{gram-pos}\}$.

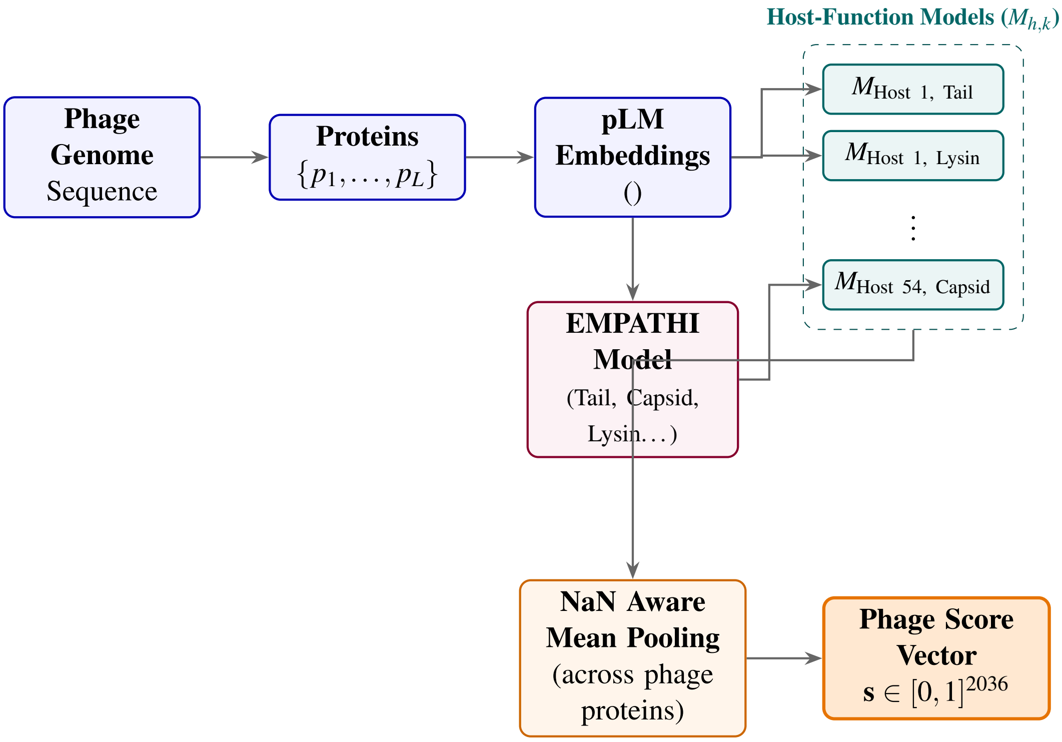

Figure 4.1: Overview of the feature extraction and score generation pipeline. Phage genomes are translated into proteins and encoded using protein language model embeddings. The EMPATHI pipeline assigns functional categories, which are evaluated across $M = 2{,}160$ host function base classifiers ($M_{h,k}$). Finally, NaN aware mean pooling aggregates protein predictions into a $K_{\text{tot}} = 2{,}036$-dimensional phage level score vector.

The dataset contains some structural missingness: approximately 30.6% to 36.5% of protein category scores are missing (NA) because any individual phage naturally only carries proteins for a subset of functional classes. An entry $s_{p,m} = \text{NA}$ indicates that protein p was not annotated with the function m, this reflects biological absence rather than random measurement noise.

Formally, we define a binary observation mask $\mathbf{M} \in \{0,1\}^{N_{\text{prot}} \times M}$ such that:

$$s_{p,m} = \text{NA} \iff M_{p,m} = 0. \tag{4.1}$$

Because these missing entries represent biological absence, imputing them with zero would falsely treat unannotated proteins as evidence against infection, which would violate the nonconformity formulation (Section 3.3). Therefore we preserve missingness via NaN aware aggregation (Section 4.3).

Set	Total Phages	Gram-Negative	Gram-Positive
Full Dataset	18,474	10,735	7,708
Test Split (Split 0)	4,433	2,811	1,611
Validation Split (Split 1)	3,173	–	–
Training Splits (Splits 2–4)	10,868	7,924	6,097

Table 4.1: Dataset Summary and Class Distributions

4.2 Protein to Phage Aggregation and NaN-Aware Pooling

The base classifiers only score individual proteins $p \in \mathscr{P}_i$, while host infection is a property of the phage as a whole, so we need to aggregate the protein level scores into a single phage-level score vector. For each phage $i \in \{1, \ldots, N_{\text{phage}}\}$ ($N_{\text{phage}} = 18{,}474$) with protein set $\mathscr{P}_i$, let $M_{p,k} = \mathbf{1}(\text{protein } p \text{ has a valid prediction for function} k)$ denotes the protein-level mask. We compute the aggregation using running sums and counts per phage, this avoids holding the full 1.85×10^6-row protein matrix in memory.

For each of the K functional dimensions, we get the aggregated phage level score using dimension wise, NaN aware mean pooling:

$$s_{i,k} = \begin{cases} \dfrac{\sum_{p \in \mathscr{P}_i} s_{p,k} \cdot M_{p,k}}{\sum_{p \in \mathscr{P}_i} M_{p,k}} & \text{if } \sum_{p \in \mathscr{P}_i} M_{p,k} > 0, \\ \text{NA} & \text{otherwise.} \end{cases} \tag{4.2}$$

Equation (4.2) excludes missing entries (`NA`) from both the numerator and denominator, and takes the average strictly over observed proteins. A score is NA if and only if no protein of phage i has an observed prediction for function k.

The phage-level mask follows directly:

$$m_{i,k} = \mathbf{1}\left(\sum_{p \in \mathscr{P}_i} M_{p,k} > 0\right). \tag{4.3}$$

i.e. $m_{i,k} = 1$ if at least one protein of phage i has an observed score for column k, otherwise, it is 0. This reduces the protein data to a phage-level matrix $\mathbf{S} \in ([0,1] \cup \{\text{NA}\})^{N_{\text{phage}} \times K}$ paired with a binary mask for missingness $\mathbf{M} \in \{0,1\}^{N_{\text{phage}} \times K}$.

As we have already established in Section 3, our conformal envelope constructions operate natively on the paired observation $(\mathbf{s}_i, \mathbf{m}_i)$. This preserves the structural missingness without fabricating values in place of it. Consequently, all conformal classification results reported in Section 5 rely entirely on this native representation, and require no imputation after aggregation. The task dependent rules that determine which protein predictions count as valid and thus define $M_{p,k}$ are detailed in Section 4.3.

4.3 Task Specific Filtering and Data Partitioning

The main requirement for the finite sample conformal validity (Theorem 2.3.1) is the exchangeability of the calibration samples S_2 and test samples S_{test}. In our dataset, the cross validation partition is defined at the phage level ($f_p \in \{0,1,2,3,4\}$), whereas the underlying base scoring models were trained at the protein-cluster (`pc`) level. A single protein cluster can span phages assigned to different genome folds, therefore a base model scoring protein p for category k may have observed homologous proteins from p's cluster during training.

Formally, let $F_{p,k} \in \{0,1,2,3,4\}$ denote the fold held out when training the base model for feature k. A naive aggregation which uses scores where $F_{p,k} \neq f_p$ introduces optimistic bias, by carrying training information into test evaluations. This breaks the exchangeability assumption that is a requirement of Theorem 2.3.1.

The underlying prediction models for host level and gram type classification differ in structure, hence we correct for data contamination at the appropriate granularity for each task:

For **host level classification**, each of the $M = 2\,160$ host function models were trained on an independent five fold split of the protein cluster space. Let $F_{p,k} \in \{0,1,2,3,4\}$ denote the held out fold identifier for protein p under the model for feature k. To prevent homology leakage across splits, we enforce an out of fold masking operator prior to phage aggregation

$$s_{p,k} \leftarrow \text{NA} \quad \text{if } F_{p,k} \neq f_p. \tag{4.4}$$

We replace the in fold predictions with `NA`, then the NaN-aware mean pooling operator (4.2) naturally excludes the contaminated scores without introducing imputation bias. This allows us to utilize the full dataset across all five folds without discarding data:

$$D_{\text{test}} = \{i : \text{split}_i = 0\}, \qquad D_{\text{val}} = \{i : \text{split}_i = 1\}, \qquad D_{\text{train}} = \{i : \text{split}_i \in \{2,3,4\}\}.$$

Here, D_{train} provides us with the the shape-discovery and size-scaling calibration sets (S_1, S_2), D_{val} is used to tune envelope hyperparameters (e.g., bin count NB, directional resolution M, and kernel smoothing κ), and D_{test} is held out for final evaluation.

For **gram type classification**, the $K = 40$ functional category models used for gram type classification do not have per column split assignment tables. As a conservative, unambiguous fallback, we restrict the gram type dataset strictly to phages in Fold 0 ($\text{split}_i = 0$). Because Fold 0 phages are consistently held out at the genome cluster level, all associated protein scores are unbiased by construction. Within this Fold 0 subset, we partition the phages $40/30/30$ into shape discovery (S_1), size-scaling (S_2), and test (S_{test}) sets following the standard CSA calibration split.

In both cases, $M_{p,k}$, which is the indicator determining which protein level scores are treated as valid input to the pooling operator of Equation (4.2), reduces to whichever correction each task's models support. This is the exact fold match test of Equation (4.4) for host level classification, or membership in the Fold-0 subset for gram type classification.

— 5 —
Experiments and Results

In this section we will present the experimental validation of the methods developed in Section 3. In section 5.1 we validate the construction of each envelope on simulated data, where the ground truth is known, we verify the finite sample coverage guarantees, geometric adaptation, and dimensionality scaling. Section 5.2 and Section 5.3 then report results on real phage data for gram-type and host-level classification respectively.

5.1 Synthetic Validation

Before applying the conformal envelope constructions of Section 3.4 to real phage data, we evaluated the theoretical guarantees, geometric adaptations and the efficiency scaling of methods on simulated scores with known ground truth.

These experiments have four specific goals:

(i) Verify the exact finite sample marginal coverage validity ($1-\alpha = 0.900$) across varying score dimensions K.

(ii) Demonstrate how class conditional calibration improves prediction set efficiency over shared global boundaries.

(iii) Characterize how non-convex geometries (Radial and Strip) adapt to non Gaussian, correlated, or sparse score distributions.

(iv) Validate the per host envelope cascade in a multi class ground truth setting.

5.1.1 2D Geometric Proof and Class Conditional Efficiency

We first simulated $N = 400$ bivariate score pairs from $\mathscr{N}((0.15, 0.15), \Sigma)$ with $\Sigma_{11} = \Sigma_{22} = 0.01$ and $\Sigma_{12} = -0.0085$, clipped to $[0,1]^2$. Fitting a single convex envelope ($\alpha = 0.10$, $M = 500$ directions, 50/50 S_1/S_2 split) confirms the two stage pipeline: the initial shape (only boundary under covers), whereas the size scaled envelope hits the target 90% coverage exactly.

Next, we introduced four classes (Hosts A–D) with centroids at $(0.1, 0.7)$, $(0.7, 0.1)$, $(0.1, 0.1)$, and $(0.7, 0.7)$, each with variance 0.01. Fitting a single shared convex envelope ($\alpha = 0.10$) to test a phage whose true host is Host B yields the prediction set $C(X) = \{\text{Host A}, \text{Host B}, \text{Host C}\}$. Because a convex hull must span the empty interior space between disjoint clusters, the prediction set remains overly conservative, even when tightening the significance level to $\alpha = 0.40$. This confirms that shared convex hull inefficiency is structural rather than an artifact of hyperparameter choice.

Computing host specific scaling thresholds $\hat{t}_h$ while sharing the directional shape $\tilde{\mathbf{q}}$ yields $\hat{t}_A = 0.969$, $\hat{t}_B = 0.980$, $\hat{t}_C = 0.302$, and $\hat{t}_D = 1.074$. The large spread in $\hat{t}_h$ reflects underlying class dispersion heterogeneity (e.g., Host C requires only 30% of the expansion required by Host D). Applying these class conditional thresholds shrinks the prediction set for the same test phage to $C(X) = \{\text{Host B}, \text{Host D}\}$, reducing set size from 3 to 2 while preserving true label coverage.

Similarly, we also evaluated the non convex Radial and Strip methods on this same 2D synthetic dataset ($K = 2$, $\alpha = 0.10$). The results confirmed that they both successfully adapt to the negatively correlated score distribution, and achieve the exact target coverage of 90.0% with and have size scaling factors of $\hat{t} = 0.996$ and $\hat{t} = 0.944$, respectively. These are consistent with the coverage results reported for all three methods in Section 5.1.2.

5.1.2 Dimensionality Scaling and Finite-Sample Coverage

To evaluate coverage validity as the dimensionality grows, we benchmarked the reference CSA paper method (binary search) against the Radial envelope ($\delta = 15^\circ$), and the Strip envelope ($M_{\text{bin}} = 20$) across dimensions $K \in \{2, 5, 10, 20\}$ at target level $1 - \alpha = 0.900$ ($\alpha = 0.10, N = 1\,000$). The scores were drawn from a K-dimensional multivariate normal distribution with mean vector $\mathbf{0.3}$, variance 0.02, and off diagonal feature correlation $\rho = 0.30$, clipped to $[0, 1]^K$.

As shown in Table 5.1, every method achieves exactly 90.0% empirical coverage across all tested dimensions. This empirically confirms the distribution free guarantee of Theorem 2.3.1. However, their statistical efficiency, which is reflected by the scaling factor $\hat{t}$, is significantly different as K increases. For the reference CSA method, the $\hat{t}$ only slightly changes from 0.996 to 1.021 between $K = 2$ and $K = 20$. For the radial method the $\hat{t}$ value inflates for high dimensions, it nearly triples from $\hat{t} = 0.972$ at $K = 2$ to $\hat{t} = 2.925$ at $K = 20$. This is due to the curse of dimensionality, the directional cones ($\delta = 15^\circ$) capture fewer local calibration points in sparse high dimensional spaces, so more directions fall back to the conservative global marginal quantile $\tilde{q}_{\text{marg}}$, inflating $\hat{t}$. Finally, the strip method maintains the robust sample efficiency (reaching only $\hat{t} = 1.212$

Dimension (K)	Reference CSA (Convex) Coverage (%)	$\hat{t}$	Radial Envelope Coverage (%)	$\hat{t}$	Strip Envelope Coverage (%)	$\hat{t}$
$K=2$	90.0	0.996	90.0	0.972	90.0	0.893
$K=5$	90.0	1.009	90.0	1.179	90.0	1.031
$K=10$	90.0	1.016	90.0	1.920	90.0	1.107
$K=20$	90.0	1.021	90.0	2.925	90.0	1.222

Table 5.1: Dimensionality scaling experiment ($K \in \{2,5,10,20\}$, nominal $1-\alpha = 0.900$, $N = 2\,000$). Empirical coverage and calibrated size scaling factors $\hat{t}$ are reported across envelope geometries.

at $K = 20$), and outperforms the radial method by using axis aligned pairwise projections.

5.1.3 Geometric Robustness Across Score Topologies

To assess the boundary adaptation under realistic non convex or misspecified feature spaces, we calibrated 3D ($K = 3$) envelopes ($\alpha = 0.10, N = 400$) across four distinct score topologies: (i) *Uncorrelated Gaussian* ($\Sigma = \mathbf{I}$), (ii) *Correlated Gaussian* ($\rho = 0.85$), (iii) *Multivariate Laplace* (heavy-tailed), and (iv) *Sparse* (one feature centered near zero, 0.05 ± 0.03, while others remain diffuse at 0.35).

Table 5.2: Geometric adaptation benchmark across 3D score topologies ($\alpha = 0.10, N = 400$). Both Radial ($M = 150$) and Strip ($M_{\text{bin}} = 12$) envelopes achieve exact nominal coverage under distributional misspecification.

Score Distribution Topology	Radial Envelope Coverage (%)	$\hat{t}$	Strip Envelope Coverage (%)	$\hat{t}$
Normal (Uncorrelated, $\Sigma = \mathbf{I}$)	90.0	1.072	90.0	1.161
Normal (Highly Correlated, $\rho = 0.85$)	90.0	0.995	90.0	0.861
Multivariate Laplace (Heavy-Tailed)	90.0	1.072	90.0	1.232
Sparse (Degenerate Dimension)	90.0	1.077	90.0	1.155

The results in Table 5.2 highlight several key properties:

- **Distribution Free Invariance:** Both Radial and Strip constructions hit the exact 90.0% target across all four topologies. this confirms that heavy tails or feature sparsity do not compromise marginal validity.

- **Adaptation to Correlation:** On highly correlated data ($\rho = 0.85$), both methods require little to no inflation ($\hat{t} \approx 1$ for radial and $\hat{t} \approx 0.861$ for strip). This indicates that the shape discovery step already produces a well fitted envelope for diagonal score clouds. The Strip method's initial conditional bounds are slightly conservative for this geometry, so the calibration step fine tunes the scale downward.

- **Heavy-Tail Sensitivity:** The heavy tailed Laplace distribution induces the largest inflation factor for the Strip envelope ($\hat{t} = 1.232$). This is because heavy tails produce extreme nonconformity values which result in higher quantile thresholds during the shape discovery step.

5.1.4 Multi-Host Benchmark Verification

Finally, we validate the per-host envelope cascade (Section 3.1) on a simulated 6-host system ($\mathscr{H} = \{\text{Host}_A, \dots, \text{Host}_F\}$, $K = 5$, $N_{\text{cal}} = 500$ infectors per host, $\alpha = 0.10$). Host-specific envelopes are calibrated from true infector score distributions centered in $[0.1, 0.5]^5$. A test phage matching Host_B (scores in $[0.1, 0.4]^5$ for Host_B; scores in $[0.6, 0.9]^5$ for all non target hosts) is evaluated against all six envelopes.

Across the reference CSA, Radial, and Strip constructions, all three methods correctly isolate $C(\mathbf{s}) = \{\text{Host}_B\}$ as an ideal singleton prediction set, excluding every non-target host. This test case serves as a sanity check under separated ground truth rather than a full coverage estimate, but it also confirms the per host cascade functions as intended prior to evaluating the substantially larger host space ($N = 54$) in Section 5.3.

5.2 Gram Type Classification

We evaluated the three envelope geometries (Radial, Strip, and Collapsed 1D) on the real phage data for a binary Gram type classification ($\mathscr{H} = \{\text{gram-neg}, \text{gram-pos}\}$). We only used the 4422 phages in split 0 to ensure that all scores remain out of fold by construction and there is no contamination.

5.2.1 Experimental Setup

We partitioned the split 0 subset into shape discovery ($|S_1| = 1\,768$), size scaling ($|S_2| = 1\,326$), and test evaluation ($|S_{\text{test}}| = 1\,328$) sets using a 40/30/30 split. For each geometry, we fitted a single shared envelope E on S_1, and then calculated the class conditional scaling factors $\hat{t}_{\text{neg}}$ and $\hat{t}_{\text{pos}}$ on S_2 using (3.9). To ensure validity across both classes simultaneously, we scaled the envelope by their maximum:

$$\hat{t} = \max\left(\hat{t}_{\text{neg}}, \hat{t}_{\text{pos}}\right). \tag{5.1}$$

At test time, we evaluated the Gram type nonconformity transformation under both candidate hypotheses. For an empty candidate set ($\emptyset$), we returned both labels ($C(\mathbf{s}) = \mathscr{H}$) as established in Section 3.5.

5.2.2 Coverage and Prediction Efficiency

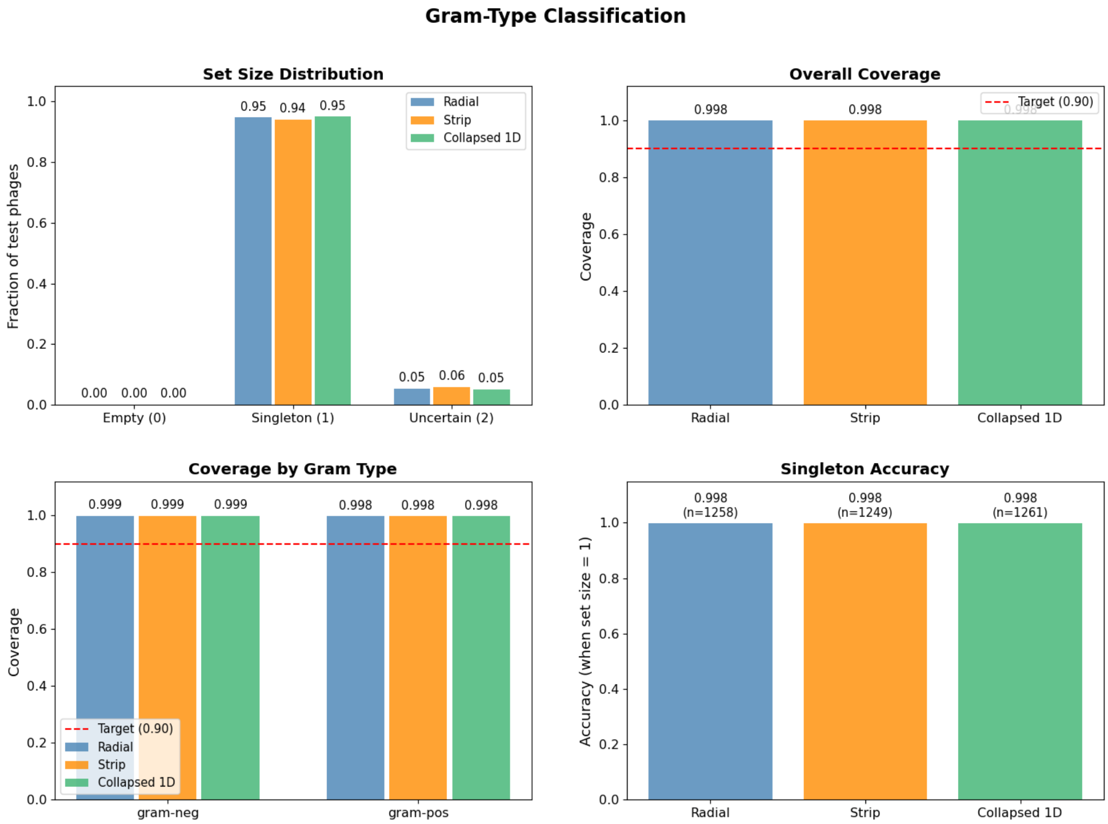

Figure 5.1: Gram type classification performance across Radial, Strip, and Collapsed 1D geometries on the test split (Fold 0, nominal target $1-\alpha = 0.90$). Panels show (left to right): prediction set size distribution, marginal empirical coverage, class-conditional coverage, and singleton accuracy with sample counts (n).

As we can see in Figure 5.1 (middle panels), all three envelope geometries achieve 99.8% empirical coverage, which is well above the nominal 90% target (red dashed line), this is a consequence of the underlying data distribution . The class conditional breakdown confirms that the coverage is balanced for both Gram negative (99.9%) and Gram positive (99.8%) phages. This demonstrates that the max threshold calibration $\hat{t} = \max(\hat{t}_{\text{neg}}, \hat{t}_{\text{pos}})$ effectively eliminates coverage disparity between classes.

We can also see that the high coverage does not compromise efficiency. The set size distribution chart of Figure 5.1 (left panel), shows that 94–95% of test phages receive singleton predictions, while only 5–6% require size 2 uncertain sets ($C(\mathbf{s}) = \mathscr{H}$). The empty set rate is 0% by construction due to the conservative fallback rule. When singleton sets are produced, the prediction accuracy is 99.8% across all three geometries ($n \geq 1\,249$), which tells us that when the model is confident enough to output exactly one label, it is almost always correct.

To check where this over coverage comes from, we evaluated coverage directly on the calibration set S_2 itself, the set on which $\hat{t}$ is calibrated. All three methods already exceed the 90% target on S_2, Radial reaches 94.9%, Strip 93.6%, and Collapsed 1D 94.5%. Since this gap appears before the envelope is applied to test data, we can say that the over coverage is a property of the calibration itself. Two factors can explain this

- the class conditional thresholds are combined using Equation (5.1), so whichever class requires the smaller threshold is scaled past its own target level by construction
- the nonconformity scores of the two classes are extremely well separated (Section 5.2.3), so the $(1-\alpha)$ quantile of the correct class already sits far below where the wrong class's scores begin, this leaves a wide margin that increases the coverage well past the target even at the calibration stage.

5.2.3 Score Separation and Geometric Equivalence

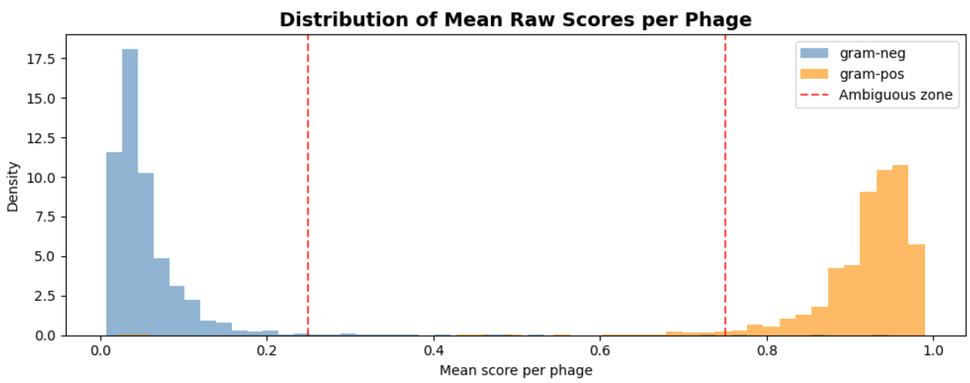

Figure 5.2: Distribution of raw scores per test phage (mean over all $K = 40$ functional category dimensions), separated by true Gram type. Gram negative phages concentrate tightly near 0 while Gram positive phages concentrate near 1, leaving the intermediate ambiguous zone (0.25–0.75, dashed red lines) almost empty.

All three envelope constructions yield nearly identical performance metrics in Figure 5.1. This equivalence can directly be explained by the empirical raw score distribution in Figure 5.2: Gram negative phages concentrate tightly near the origin (mean 0.055 ± 0.054), whereas Gram positive phages concentrate near 1.0 (mean 0.918 ± 0.066). The two classes form disjoint, well separated clusters with negligible density in the ambiguous zone (0.25–0.75), any sensible star shaped boundary cleanly separates them.

This near identical performance across the three geometries suggests that the choice of envelope geometry is less critical for low dimensional task with strongly separated scores. The advantages of non convex can be seen more prominently in the higher dimensional, multi class host level classification task in Section 5.3

5.3 Host Level Classification

Host level prediction extends our conformal prediction framework from binary attributes to a multiclass task across $N = 54$ candidate bacterial hosts ($\mathscr{H} = \{h_1, \dots, h_{54}\}$). This setting introduces statistical and dimensional challenges: heavy label imbalance, small per host calibration samples, and inference through a hierarchical two-stage cascade. Rather than attempting to fit a 2036 dimensional global envelope across the joint feature space, we constructed class conditional envelopes which were calibrated independently for each candidate host $h \in \mathscr{H}$ on its dedicated K_h dimensional score sub vectors $\mathbf{s}_h$.

5.3.1 Experimental Protocol

We apply the split conformal framework of Section 3 to the reconstructed score vectors (3.5) of the $N_{\text{tot}} = 15{,}638$ annotated phages across 54 candidate hosts ($K_h \in [26, 40]$ per host; $K_{\text{tot}} = 2{,}036$). We again use a three way partition of the genome cluster data:

- D_{train} (Split 2–4, $N_{\text{train}} = 9{,}101$): For each candidate host with sufficient training data ($n_h \geq 3$), the infector phages are partitioned 50/50 into shape-discovery ($S_1^{(h)}$, $\lfloor n_h/2 \rfloor$) and size-scaling ($S_2^{(h)}$, $\lceil n_h/2 \rceil$) subsets to fit E_h and compute $\hat{t}_h$.
- D_{val} (Split 1, $N_{\text{val}} = 2{,}593$): used for envelope hyperparameter tuning.
- D_{test} (Split 0, $N_{\text{test}} = 3{,}944$): held out and evaluated once after fixing all parameters.

Unlike Gram type classification (Section 5.2), host level scores use per model split assignments (Section 4.3) to ensure the out of split validity across all five splits rather than restricting calibration to split 0 alone. Missing entries (NA) are handled by each envelope geometry without data imputation

At test time, the Gram type prediction set restricts the candidate hosts to $\mathscr{H}_{\text{cand}}(s, m)$. The host prediction set $C_{\text{host}}(s, m)$ is evaluated via Equation 3.1. If a test phage falls outside all candidate envelopes ($C_{\text{host}}(s,m) = \emptyset$), we get an empty set which is resolved by selecting the candidate which minimizes the normalized nonconformity:

$$h_{\text{forced}}(s) = \arg \min_{h \in \mathscr{H}_{\text{cand}}(s,m)} \frac{\tau_h(s_h)}{\hat{t}_h}, \tag{5.2}$$

which selects the closest boundary without defaulting to the uninformative full label set.

5.3.2 Hyperparameter Selection on Validation Data

We optimized the hyperparameters on D_{val} (Split 1) by minimizing the average set size subject to empirical coverage constraints at target level $1-\alpha = 0.90$:

$$\theta^\star \in \arg\min_{\theta} \overline{|C|}_{\text{val}}(\theta) \quad \text{subject to} \quad \hat{c}_{\text{val}}(\theta) \geq 1-\alpha. \tag{5.3}$$

with ties or near constraint cases resolved by minimizing the coverage discrepancy $|\hat{c}_{\text{val}}(\theta) - (1-\alpha)|$.

Validation under the unrestricted candidate space selected:

- **Strip Binning (N_B):** Swept over $N_B \in \{3,5,8,10,15,20\}$ with windowed monotonic smoothing ($w = 2$). $N_B = 10$ was selected, achieving 90.5% validation coverage with an average prediction set size of 12.97 candidate hosts.
- **Radial Resolution (M, κ):** Swept across $M \in \{100,250,500\}$ directions and concentration parameters $\kappa \in \{4.0,8.0,16.0\}$. $M = 100$ with $\kappa = 4.0$ provided the optimal trade-off closest to nominal validity (89.0% validation coverage, average set size 12.4).

5.3.3 Empirical Test Results

Table 5.3 and Figure 5.3 summarize the empirical performance metrics evaluated on the test split ($N_{\text{test}} = 3{,}944$ phages) across all 54 hosts.

Method	Coverage	Avg. Set Size	Singleton Rate	Singleton Acc.	Forced Rate
Radial Envelope ($M = 100, \kappa = 4$)	0.939	15.25	0.131	0.896	0.023
Strip Envelope ($N_B = 10, w = 2$)	0.905	15.35	0.007	0.321	0.002
Collapsed 1D	0.929	6.09	0.286	0.937	0.014

Table 5.3: Performance metrics on the test split

As demonstrated in Table 5.3 and Figure 5.3, all three geometries achieve empirical coverage close to the 90.0% target. While the binary Gram setting produced an over conservative coverage (99.8%) due to global max threshold, the independent per host envelope calibration prevents over coverage (90.5%–93.9%).

Prediction efficiency diverges sharply:

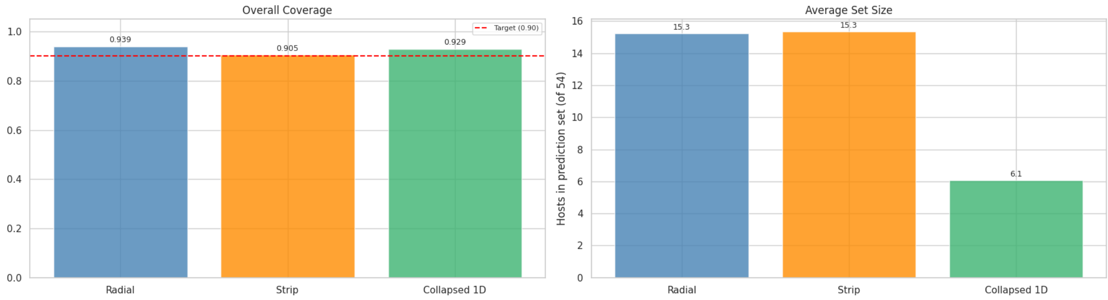

Figure 5.3: Host classification performance on Split 0 test phages ($1-\alpha = 0.90$). Left panel: overall empirical coverage compared against the target (red dashed line). Right panel: average prediction set size across candidate hosts.

(i) **Efficiency of Collapsed 1D:** Collapsed 1D achieves an average prediction set size of 6.09, which is less than half that of Radial (15.25) or Strip (15.35). In moderate to high dimensions ($K_h \geq 26$), 1D mean pooling avoids the sample sparsity penalties that arise due to angular cones and multi dimensional bins. The complete per host set size distributions and boxplots across all 54 hosts are in Appendix A Figure figs. A.3 to A.6

(ii) **Singleton Accuracy:** Collapsed 1D produces a single host for 28.6% of test phages and achieves 93.7% accuracy on these singletons. In contrast, Radial produces singletons for only 13.1% of test phages (with 89.6% accuracy), while Strip almost never gives a singleton output (0.7%).

(iii) **Forced Decision Rate:** The forced fallback rule (5.2) was only used for 1.4% of test phages for Collapsed 1D and 2.3% for Radial. This confirms that host specific envelopes reliably capture empirical infector score distributions.

Figure figs. A.1 to A.2 in the Appendix A illustrates the class conditional validity across all hosts. The coverage remains well calibrated across major host types (e.g., *Escherichia*, *Pseudomonas*, *Staphylococcus*, and *Mycobacterium*), but we can also find low coverage to the host classes with sparse data.

5.3.4 Geometric Boundary Visualization Across All Methods

In sections 5.3.3 we have seen the aggregate metrics report which tell us *how much* the three envelope construction methods differ in coverage and set size, but not *why*. To illustrate the structural mechanisms resulting in these differences, Figure figs. 5.4 to 5.5 displays the calibrated boundary projections on a two dimensional subspace (built only using two of the functional dimensions while holding the values on the remaining $K_h - 2$

dimensions fixed). Therefore, the boundary shown is just a 2D a slice of the true K_h dimensional envelope.

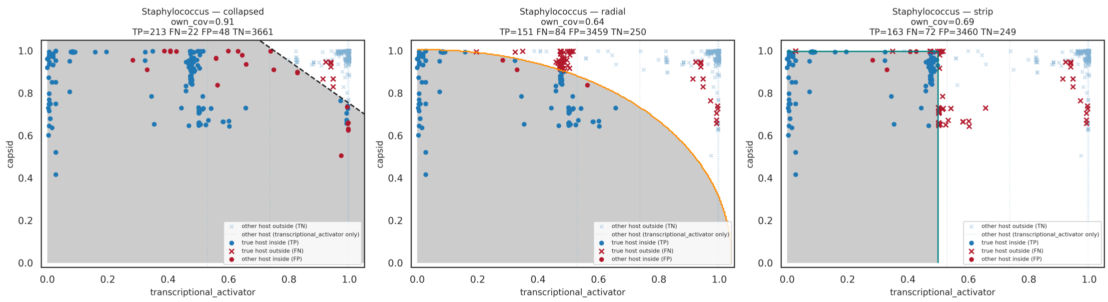

Figure 5.4: **2D Projection of Calibrated Boundaries on Good Predictors.** Per host envelope boundary comparison for Staphylococcus projected onto the 2D plane of two discriminative functional features (transcriptional_activator vs. annealing). Left (Collapsed 1D): Linear boundary defined by equal weight mean nonconformity. Centre (Radial): Smooth angular kernel boundary ($\kappa = 4.0$), closely enveloping the nonconformity cluster. Right (Strip): Piecewise constant staircase boundary enforcing local monotonicity ($w = 2$).

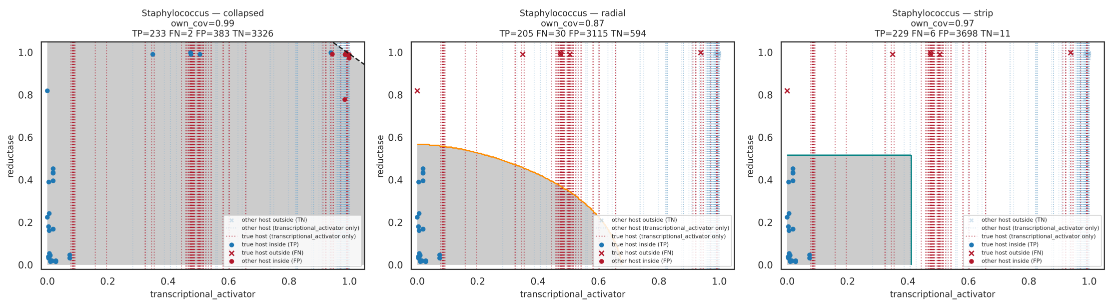

Figure 5.5: **2D Projection of Calibrated Boundaries on One Good and One Bad Predictor**. Per host envelope boundary comparison for Staphylococcus projected onto one informative and one uninformative functional feature (transcriptional_activator vs. capsid) When one dimension contains non infector noise, Collapsed 1D's linear projection dilutes the strong signal across both axes. The Strip envelope isolates the noisy axis to wide marginal bounds along the horizontal strip, while the Radial envelope expands outward along the uninformative ray.

Collapsed 1D's boundary is the straight anti diagonal line implied by its scalar mean threshold (Section 3.4.1). A phage that is strong in lysin but weak in annealing is treated identically to one with the reverse profile, as long as the average clears the threshold.

This produces its smallest false positive count of the three (21, `own_cov` $= 0.94$), but at the cost of giving us 15 false negatives. The true infectors whose evidence concentrates in one dimension rather than both are missed by a boundary which is blind to that asymmetry.

The boundary of the Radial method is the smooth arc produced by the radius of Equation (3.7). It curves inward for phages strong in one dimension but weak in the other. This recovers more true infectors (10 false negatives, vs. Collapsed's 15) but at the cost of efficiency: 2,568 false positives, since a boundary calibrated from points concentrated near the axes extends generously along both axes individually. So the higher dimensional inflation behavior anticipated in Section 5.1.2's synthetic K-sweep, can now be visualized in real data.

Strip's staircase boundary sits between the two: it tracks the true cluster's shape more tightly than Collapsed's line, and without the smooth over extension of Radial's arc. It achieves the fewest false negatives of the three (5) but at the cost of the most false positives (2,540). It has piecewise constant conditioning (Section 3.4.3), so it captures the cluster's shape well locally, but each bin transition risks admitting points just past a true infector's actual boundary.

Across both functional feature projections, the geometric trade offs are consistent: Collapsed 1D enforces a global linear trade-off that produces tight rejection regions at the cost of directional blindness; the Radial envelope smoothly hugs directional density but expands along sparse orthant rays in higher dimensions; and the Strip envelope captures non linear conditional dependencies along dense axes via its staircase structure, but becomes overly permissive in sparse bins due to marginal fallback quantile padding.

Figure 5.6 visualizes the global inclusion topology of the Collapsed 1D predictor across the test set ($N_{\text{test}} = 3{,}944$). The clustermap confirms the gram type seperationion: Gram positive phages strictly activate Gram positive host columns (e.g., *Mycobacterium*, *Gordonia*, *Streptomyces*) with zero cross gram type false positive inclusions

The clustermaps for the Radial and Strip envelopes are provided in Appendix A.4 (Figures A.10 and A.11), these visually demonstrate how high dimensional sparsity forces broad, uninformative multi host inclusion blocks across the entire Gram negative sub matrix.

5.3.5 Effect of Gram-Type Filtering

Table 5.4 compares the performance of of the three methods with and without Gram type pre filtering.

Gram type filtering produces no change in the average set size for the collapsed 1D method which shows that the 1D mean score already provides strong discrimination between gram types. But it delivers a significant efficiency gain for both the multidimensional envelopes. The average prediction set size decreases by over 60% for both of

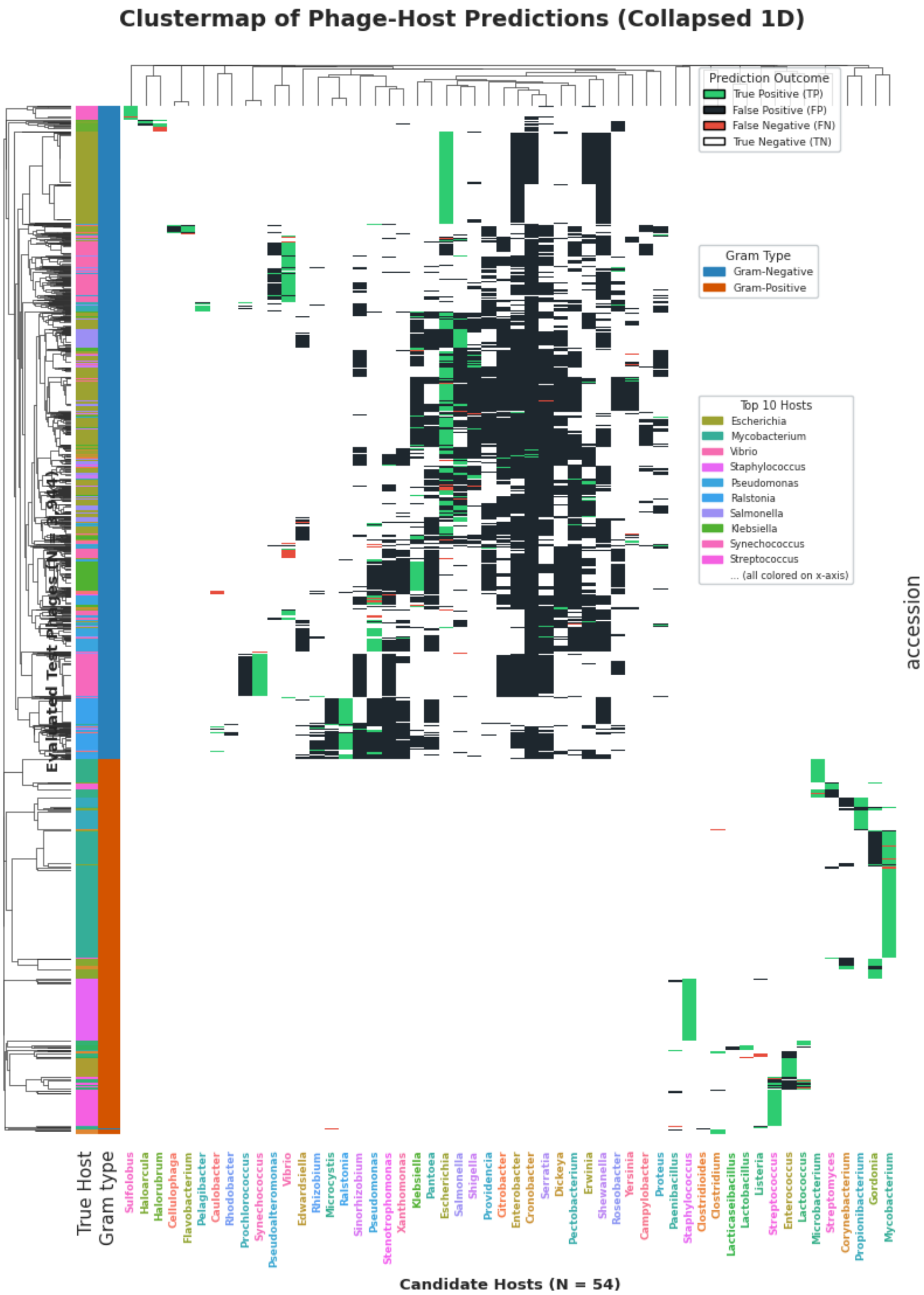

Figure 5.6: **Clustermap of Predicted Phage Host Interactions (Collapsed 1D).** Binary prediction matrix evaluated across all $N_{\text{test}} = 3{,}944$ test phages (rows) and 54 candidate host genera (columns) under the Collapsed 1D method ($1 - \alpha = 0.90$). Dark pixels denote host inclusion in the conformal prediction set $C(s)$. Left sidebars indicate ground-truth Gram phenotype (blue: Gram-negative, orange: Gram-positive) and top true host genera.

Method	Configuration	Coverage	Avg. Set Size
Radial Envelope	Unrestricted (Full $\mathscr{H}$, $N = 54$)	0.923	39.22
	Gram-Type Filtered ($\mathscr{H}_{\text{cand}}$)	0.939	15.25
Strip Envelope	Unrestricted (Full $\mathscr{H}$, $N = 54$)	0.905	39.35
	Gram-Type Filtered ($\mathscr{H}_{\text{cand}}$)	0.905	15.35
Collapsed 1D	Unrestricted (Full $\mathscr{H}$, $N = 54$)	0.924	6.09
	Gram-Type Filtered ($\mathscr{H}_{\text{cand}}$)	0.929	6.09

Table 5.4: Impact of Gram type filtering on host prediction.

the Radial ($39.22 \to 15.25$) and the Strip envelope ($39.35 \to 15.35$) and the empirical coverage also improves slightly.

5.3.6 Functional Dimension Sufficiency

The high dimensionality of the host level score space ($K_h \in [26, 40]$ per host) raises a natural question: how many of these dimensions are actually necessary? Here we checked if a single subset of functional categories, retained across all 54 hosts, can reproduce the performance of full dimensional. We selected the `TOP_N_CORE` most informative functional categories. To select the top functions, for each host we ranked its functional dimensions by their mean score on the phages that truly infect the host in the training set, then took the top 5 ranked functions per host and then finally, we pooled the top 5 list across the 54 hosts and selected the top 7 function names that occured most frequently. This yielded a core score matrix of 356 columns (17.5% of the original $K_{\text{tot}} = 2{,}036$).

Figure 5.7 summarizes the results. Collapsed1D's coverage falls slightly from 0.929 to 0.905 (still above the target) with a modest increase in average set size ($6.09 \to 6.43$). Radial's coverage falls similarly ($0.939 \to 0.915$) while its average set size *decreases* ($15.25 \to 14.58$). Strip, at its adjusted α, achieves both higher coverage ($0.905 \to 0.917$) and a smaller average set size ($15.35 \to 13.64$), the largest efficiency gain of the three methods.

These results suggest that most of the discriminative signal in the 40-dimensional score space is concentrated in a relatively small subset of functional categories, rather than being spread evenly across all of them. Retaining just 7 of the 40 categories (17.5%), which means 356 of the $K_{\text{tot}} = 2{,}036$ total host function columns preserves nearly all of the predictive performance. This indicates that most discriminative functional evidence concentrates in a compact subspace of core categories, and gives us empirical motivation for the learned low dimensional projections proposed in Section 6.3.

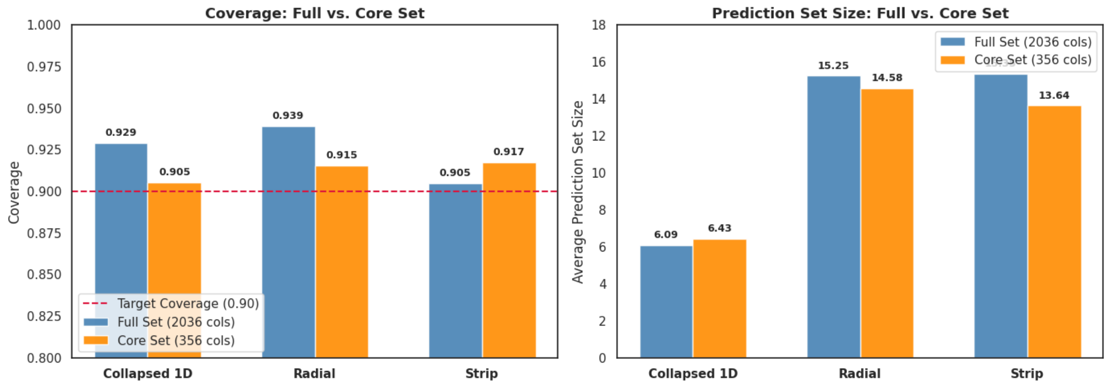

Figure 5.7: Coverage (left) and average prediction set size (right) on the full (2,036-column) vs. core (356-column) functional score set, across all three envelope geometries.

5.3.7 Performance under Different Constraints

We conducted three targeted experiments to evaluate specific operational boundaries:

- **Low Data Host Sensitivity:** Restricting evaluation to the 17 host genera with ≥ 100 calibration phages increases the Strip coverage from 0.905 to 0.969. This confirms that the slight under coverage of the strip method is driven primarily by low sample hosts with poorly estimated bins. Bu this ablation also reduces the Strip bin count from $N_B = 10$ to $N_B = 5$ in the same step, so part of the observed gain may be due to coarser binning rather than host sample size alone.
- **Dimensionality Reduction:** Restricting host subspaces to their top 3, 5, or 8 most informative functional dimensions degraded empirical coverage to 86–88%. This indicates that secondary functional annotations provide discriminative signal rather than noise.
- **Two-Dimensional Functional Subspaces:** Evaluating phages using only lysin and tail fiber dimensions (available for 51 of 54 hosts) yielded coverages of 0.857, 0.910, and 0.928 for Collapsed 1D, Radial, and Strip, respectively. Strip outperforms other geometries in this low dimensional regime, where its 2D grid avoids the curse of dimensionality.

In an alternative calibration scheme in which we incorporated non infector scores, we evaluated to bound false positives. However, penalizing non infector overlap resulted in over constraining, resulting in prediction abstention ($|C(s)| = 0$) on $> 99.9\%$ of test phages. This invalidates conformal coverage guarantees, therefore we will document it as an open methodological challenge in Section 6.3

— 6 —

Conclusion and Future Work

6.1 Summary

Existing computational phage host prediction tools output point predictions or scalar scores, which offer no finite sample guarantees that the true host is contained within the predicted candidates. This report addresses this limitation by developing a distribution free conformal prediction framework that is tailored to high dimensional, sparse protein score spaces. We adapted and extended the Conformal Set Aggregation (CSA) technique to classification, by replacing the standard convex envelopes that become unnecessarily large when class clusters are separated

The main aspects of this work are summarized as follows:

(i) **Non Convex Envelope Geometries:** We introduced two non convex geometries (Radial and Strip) alongside a Collapsed 1D mean pooling envelope. All three envelope methods met the 90.0% target coverage (90.5%–93.9%) on the test split ($N_{\text{test}} = 3\,944$ phages across 54 host genera).

(ii) **Closed Form Analytical Calibration ($\tau(\mathbf{s})$):** We derived exact, closed form expressions for the per point inclusion scaling function $\tau(\mathbf{s})$ across all three geometries. This eliminates the iterative numerical binary search required by standard CSA, which reduces the calibration complexity on the scaling set S_2 from $\mathcal{O}(|S_2| \cdot N_{\text{iter}})$ to a much better $\mathcal{O}(|S_2| \log |S_2|)$ one pass sort.

(iii) **Class Conditional Host Calibration:** Rather than computing a single global threshold, our framework calibrates class conditional quantiles $\hat{t}_h$ independently per host. This ensures finite sample coverage guarantees for individual host classes regardless of severe dataset imbalance (ranging from $> 1\,400$ phages down to single-digit counts).

(iv) **Multi Stage Empirical Validation:**

- *Synthetic Tests:* Confirmed exact $1-\alpha$ coverage across varied score topologies (Gaussian, correlated, heavy tailed, sparse) while exposing dimensional bottlenecks (Radial scaling factors $\hat{t}$ inflated with dimension K, whereas Strip remained robust).
- *Gram Type Classification:* Achieved 99.8% coverage and 94–95% singleton efficiency across all geometries due to strong underlying score separation.
- *Host Classification:* Exposed clear geometric trade offs under high dimensionality ($K_h \geq 26$). Collapsed 1D delivered the tightest sets (mean size 6.09 vs. 15.25 for Radial and 15.35 for Strip) and isolated single definitive hosts for 28.6% of test phages at 93.7% precision.

(v) **Gram to Host Cascading:** We proved finite sample validity under cascade composition (Proposition 3.1.1). We pre filtered hosts first by Gram type conformal set which pruned the candidate spaces effectively without compromising empirical coverage.

In summary, this framework provides exact coverage guarantees for phage host prediction using fast, closed form inference. It offers a practical trade off between compact prediction sets and directional sensitivity to fit different application needs.

6.2 Limitations

Despite the empirical results, this framework has several methodological and biological limitations:

- **Small Sample Host Instability:** Training sample sizes vary widely across hosts (ranging from $> 1\,400$ phages for *Mycobacterium* down to fewer than 10 for rare genera such as *Sulfolobus* and *Microcystis*). Abundant hosts calibrate reliably near the 90% target, but low sample hosts show significantly higher empirical variance, since the quantile estimates on small calibration sets ($n_h < 10$) struggle to capture the tails of the true infector score distribution. If we restrict our evaluation to hosts with ≥ 100 training phages, we can recover coverage for the Strip method (5.3.7), but this ablation also reduced the bin count from the validation-tuned $N_B = 10$ to $N_B = 5$ in the same step, so we cannot attribute the improvement just to sample size.
- **High Dimensional Sensitivity:** Collapsed 1D achieves compact prediction sets by compressing score vectors into a scalar mean, making it unable to distinguish a phage whose evidence concentrates in one informative dimension from one whose evidence is spread thinly across many uninformative ones (Fig 5.5). Conversely,

the Radial and Strip envelopes preserve this directional structure but suffer as K_h grows toward 40: sparse angular or bin-wise support forces frequent fallback to the conservative marginal quantile, which is the direct cause of their larger average prediction sets relative to Collapsed 1D.

- **Structural Missingness:** Phages frequently lack entire functional protein classes, producing $30\% - -36\%$ missing values per score vector. The Strip envelope handles these missing coordinates by removing the corresponding dimensions from the pairwise construction. Consequently, only pairs (j,i) for which both dimensions are observed contribute to the Strip scaling function.

- **Large Absolute Set Sizes:** Even where coverage is valid, the prediction sets of Radial and Strip are on average of size 15+ out of 54 candidate hots. A list of that size is of limited direct use as a clinical or laboratory shortlist, even though it is a large efficiency gain over the unrestricted, non Gram filtered setting (avg. size ≈ 39, Table 5.4).

- **Error Propagation:** Because host level prediction depends on Gram type filtering, the theoretical miscoverage is added ($\alpha + \max_h \alpha_h$, Proposition 3.1.1). Although the empirical Gram type miscoverage was near zero in our evaluation, any systematic bias in the first stage classification, (for example, on phages with ambiguous raw scores near the 0.25–0.75 zone) prunes the true host before the host level envelope is even evaluated, and this cannot be corrected.

- **Conservativeness of Strip Envelopes in High Dimensions:** The strip envelope calibrates each individual limit (each (j,i) constraint) at the $1-\alpha$ quantile level before taking their intersection. Every direction independently discards an α fraction of the data, therefore their intersection compounds these discarded data. This makes the shape discovery envelope overly tight as dimensions increase. So the the size scaling factor $\hat{t}$ is forced to grow much larger than necessary to restore target coverage.

6.3 Future Work

The challenges identified in 6.2 point toward several avenues for future research:

- **Reducing Prediction Set Sizes Directly:** Beyond the Gram type filter we already used, we can use a second coarser filtering stage, for example something based on phage morphology or a nearest neighbor prefilter could narrow the host level candidate space even further before the conformal envelope test is applied.

- **Pooling for Low Sample Hosts:** Rather than calibrating all of the 54 host envelopes independently, a hierarchical shrinkage scheme could pool information across related hosts. Shrinking quantile estimates for data scarce genera ($n_h < 10$) toward family or Gram level baselines would stabilize coverage without requiring additional data.

- **Dimensionality Reduction:** To resolve high dimensional sparsity in multi dimensional geometries ($K_h \approx 40$), future work should explore learning host specific linear or non linear projections $\Phi_h : \mathbb{R}^{K_h} \to \mathbb{R}^d$ ($d \in [2,5]$) designed to maximize infector/non infector separation.

- **Specificity Control:** The impostor experiment (5.3.7) shows that naively incorporating non infector scores collapses the prediction sets to near total abstention. Future work may investigate two sided nonconformity functions that jointly penalize false negatives and false positives, this may bound the false discovery rate and yet avoid the coverage loss.

- **Missingness Aware Envelope:** The current Strip construction handles structural missingness by removing unobserved dimensions from the pairwise envelope. Future work could investigate whether the observed missingness pattern itself contains some information about the host and if it could be used to get a missingness aware conditional envelope construction.

- **Dimensionality Reduction** As demonstrated by our analysis of core functions (Section 5.3.6), most discriminative information lives in a low dimensional subspace. In higher dimensions ($K_h \approx 40$), sample sparsity forces directional cones and grid bins to fall back onto conservative marginal bounds. Future work should explore linear dimensionality reduction such as class specific Principal Component Analysis (PCA) to extract dominant functional variance directions. Supervised linear/non-linear projections $\Phi_h : \mathbb{R}^{K_h} \to \mathbb{R}^d$ ($d \in [2,5]$) optimized to maximize infector/non infector separation. Calibrating Radial and Strip envelopes within these compact, orthogonal PCA subspaces would eliminate the curse of dimensionality while preserving directional sensitivity/

- **Per Direction Quantile Level:** For the conservativeness of strip envelopes, a next step is to calibrate each individual direction at a stricter quantile level of $1-\alpha/d$ (where d is the number of dimensions). This would ensure each that direction only removes a small of data. This would allow the final intersection to match the target $1-\alpha$ coverage, and produce a $\hat{t}$ that is closer to 1 and is less conservative.

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Host Classification Results

A.1 Class Conditional Coverage Breakdown

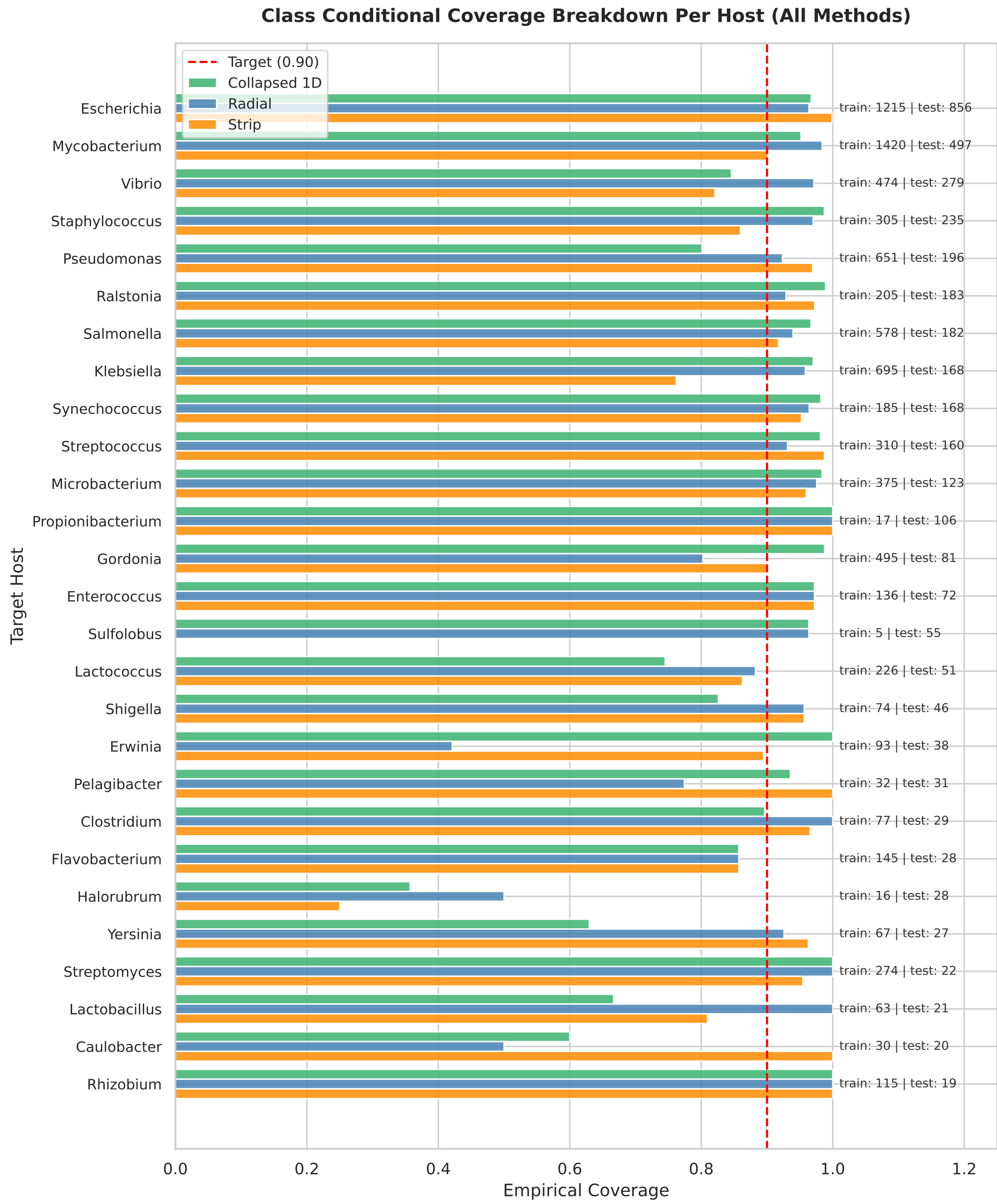

Figure A.1: Class conditional empirical coverage across all 54 candidate bacterial hosts on Split 0 test phages. The red dashed line denotes the nominal $1 - \alpha = 0.90$ target.

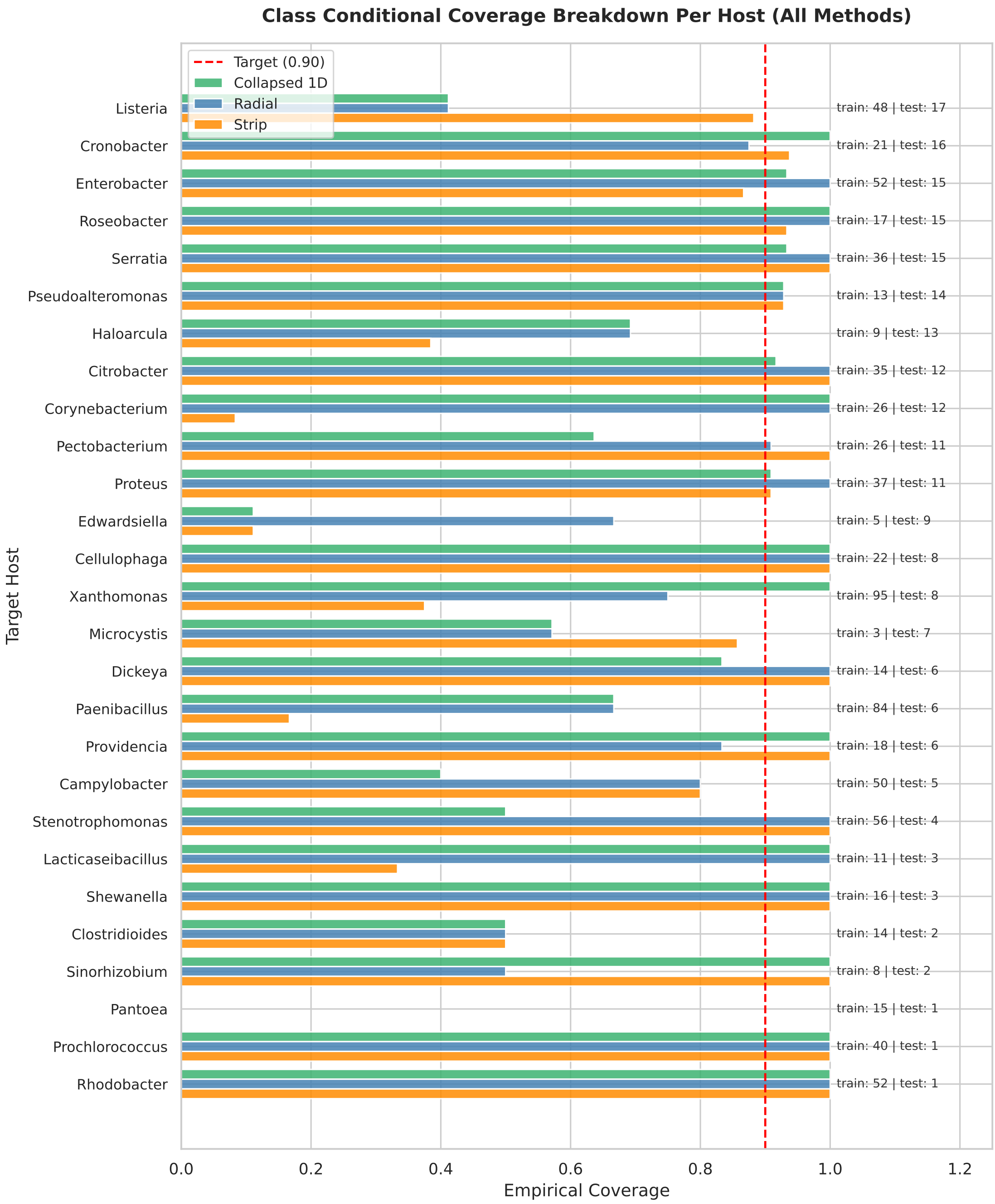

Figure A.2: Class conditional empirical coverage across all 54 candidate bacterial hosts on Split 0 test phages. The red dashed line denotes the nominal $1-\alpha = 0.90$ target.

A.2 Set Size Distribution Per Host

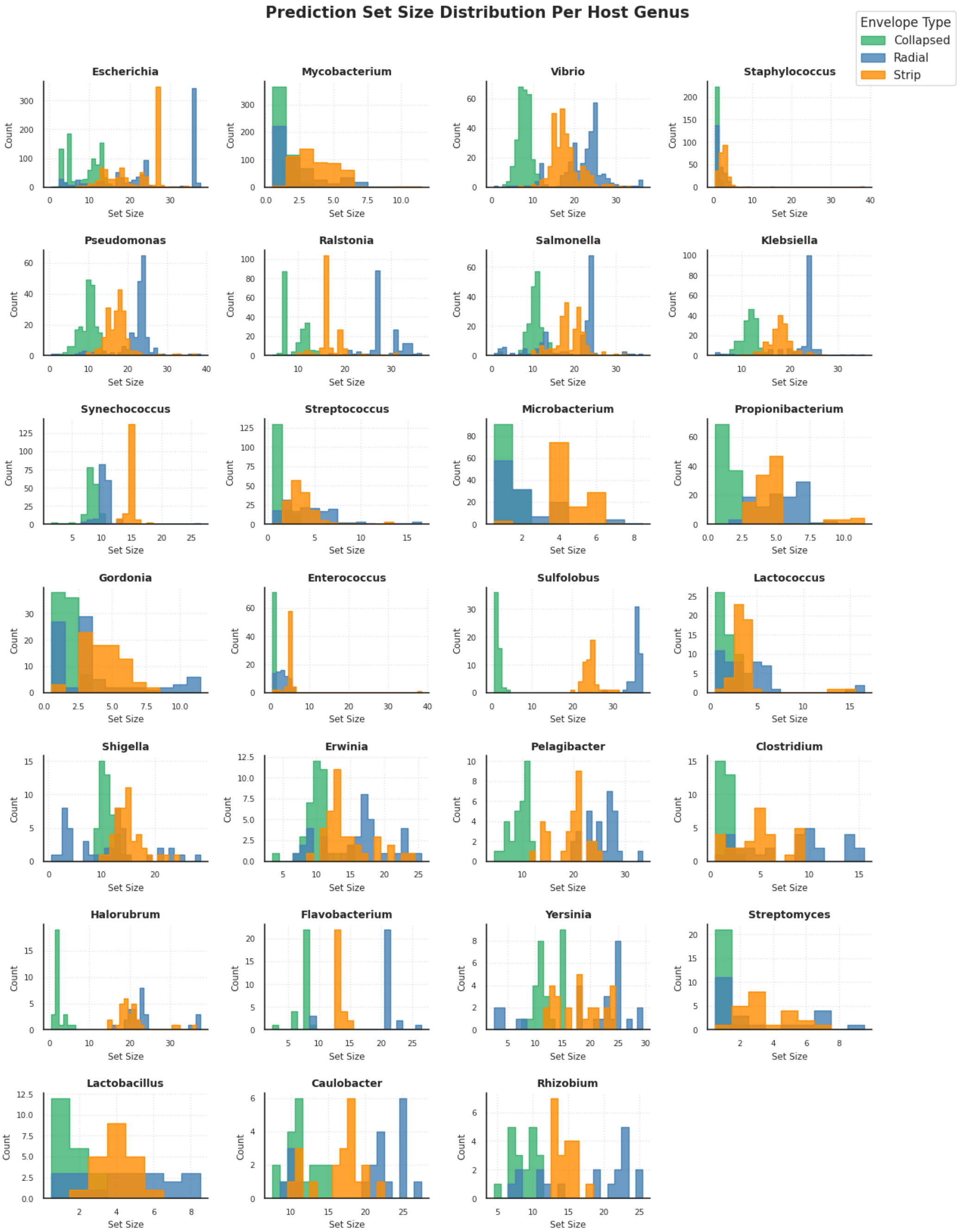

Figure A.3: Prediction set size histograms for all 54 host types across Collapsed 1D, Radial, and Strip envelopes.

Figure A.4: Prediction set size histograms for all 54 host types across Collapsed 1D, Radial, and Strip envelopes.

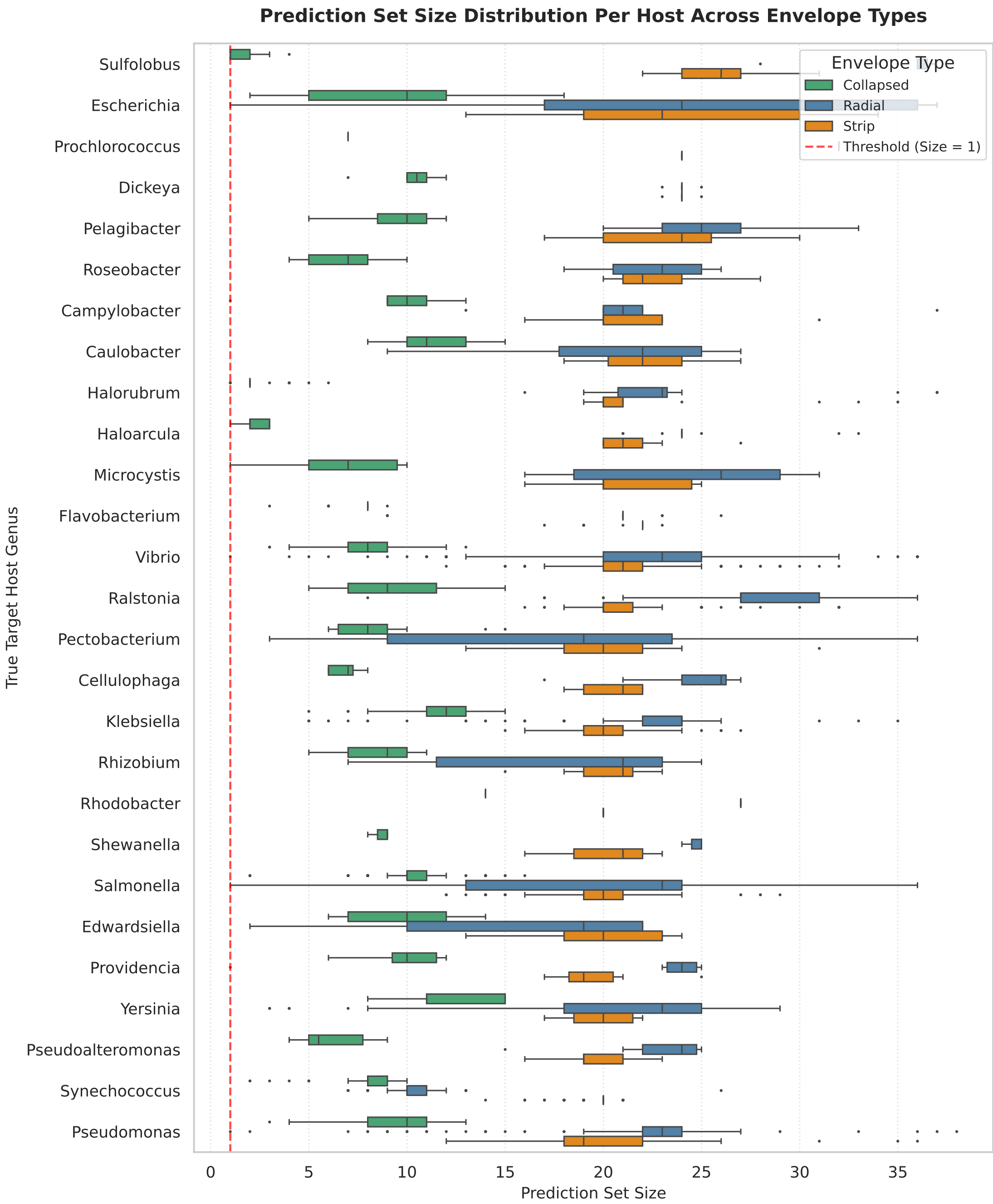

Figure A.5: Prediction set size histograms for all 54 host types across Collapsed 1D, Radial, and Strip envelopes.

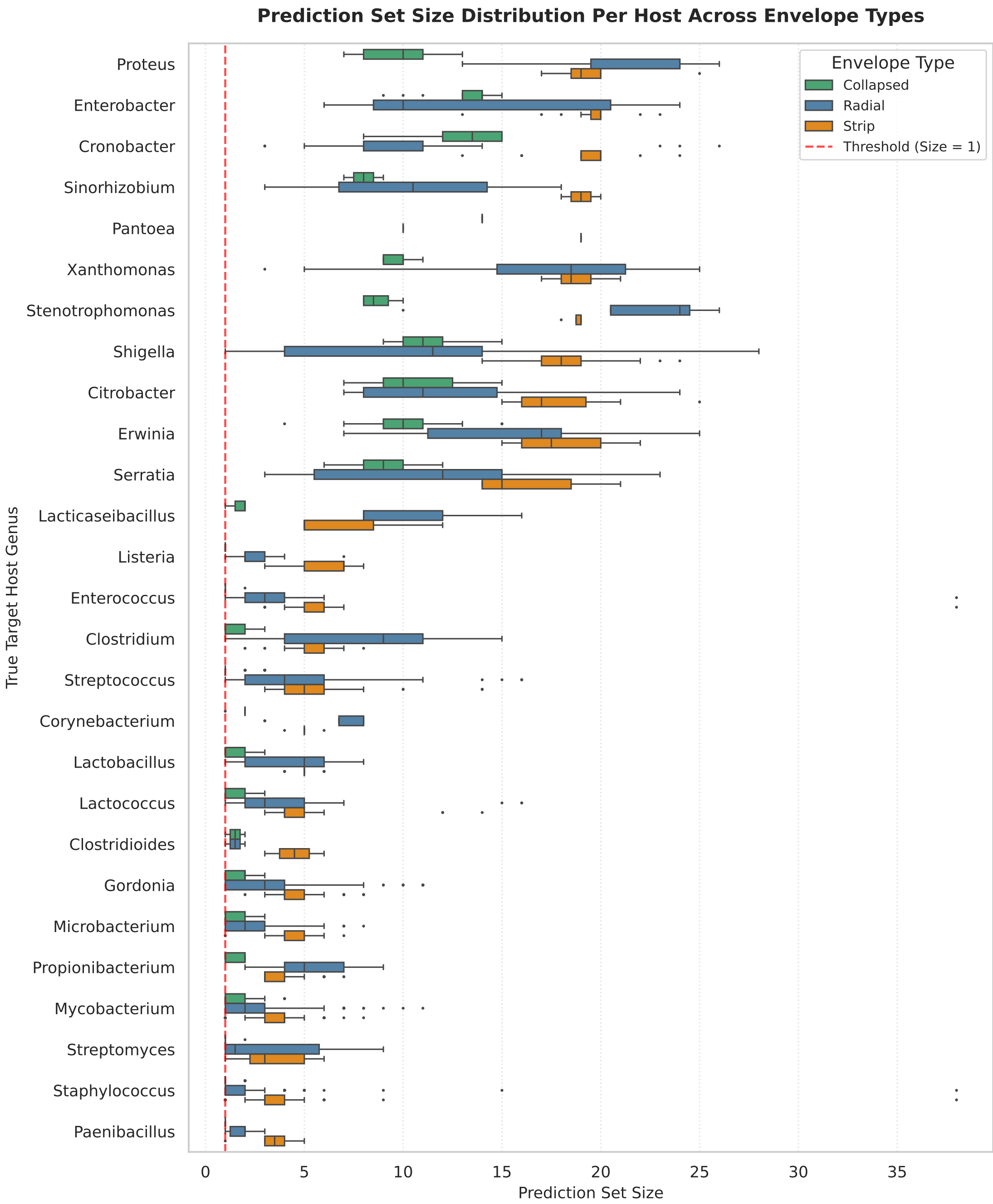

Figure A.6: Prediction set size histograms for all 54 host types across Collapsed 1D, Radial, and Strip envelopes.

A.3 Prediction Set Inclusion Heatmaps

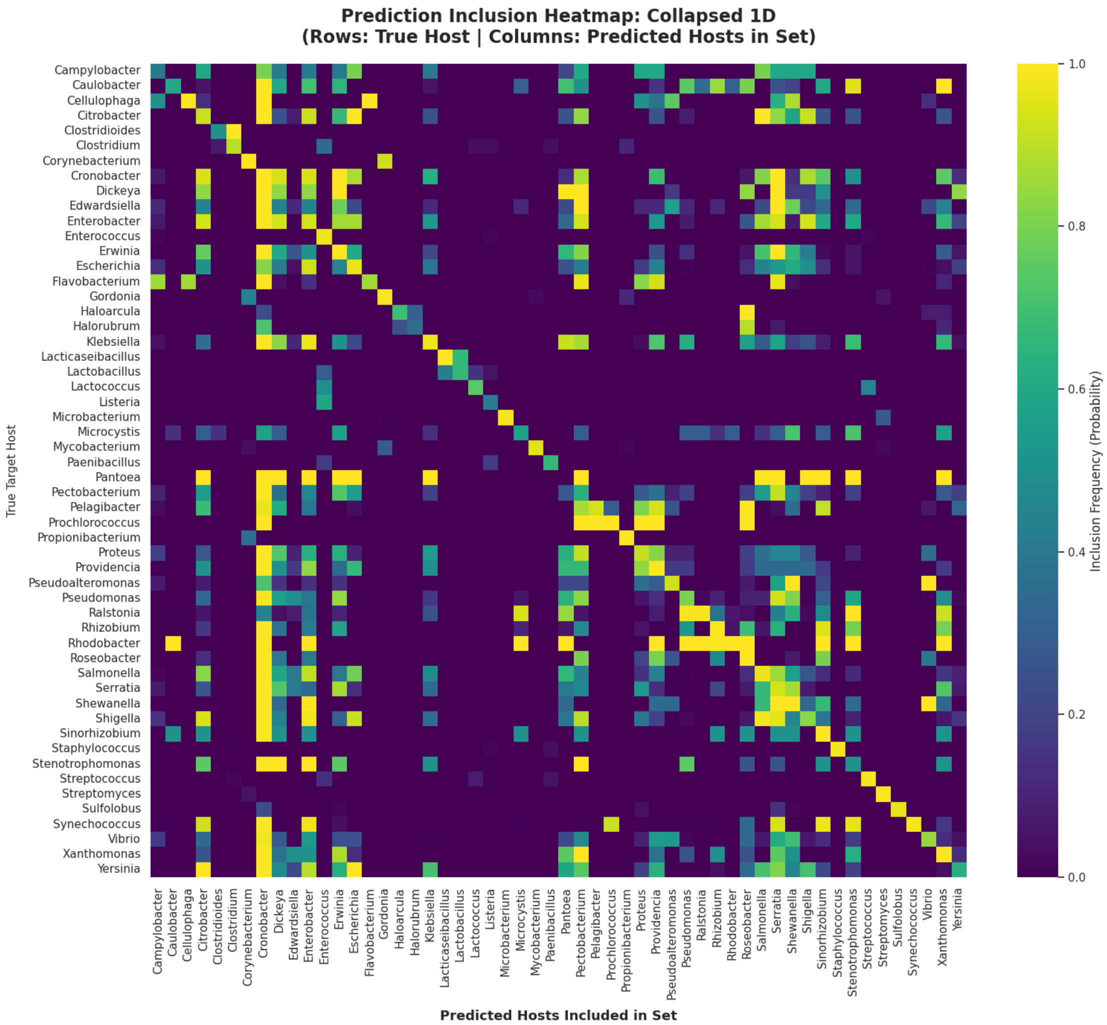

Figure A.7: Prediction set inclusion frequency heatmap for the 1D envelope across all 54 true hosts (rows) vs. predicted candidate hosts (columns).

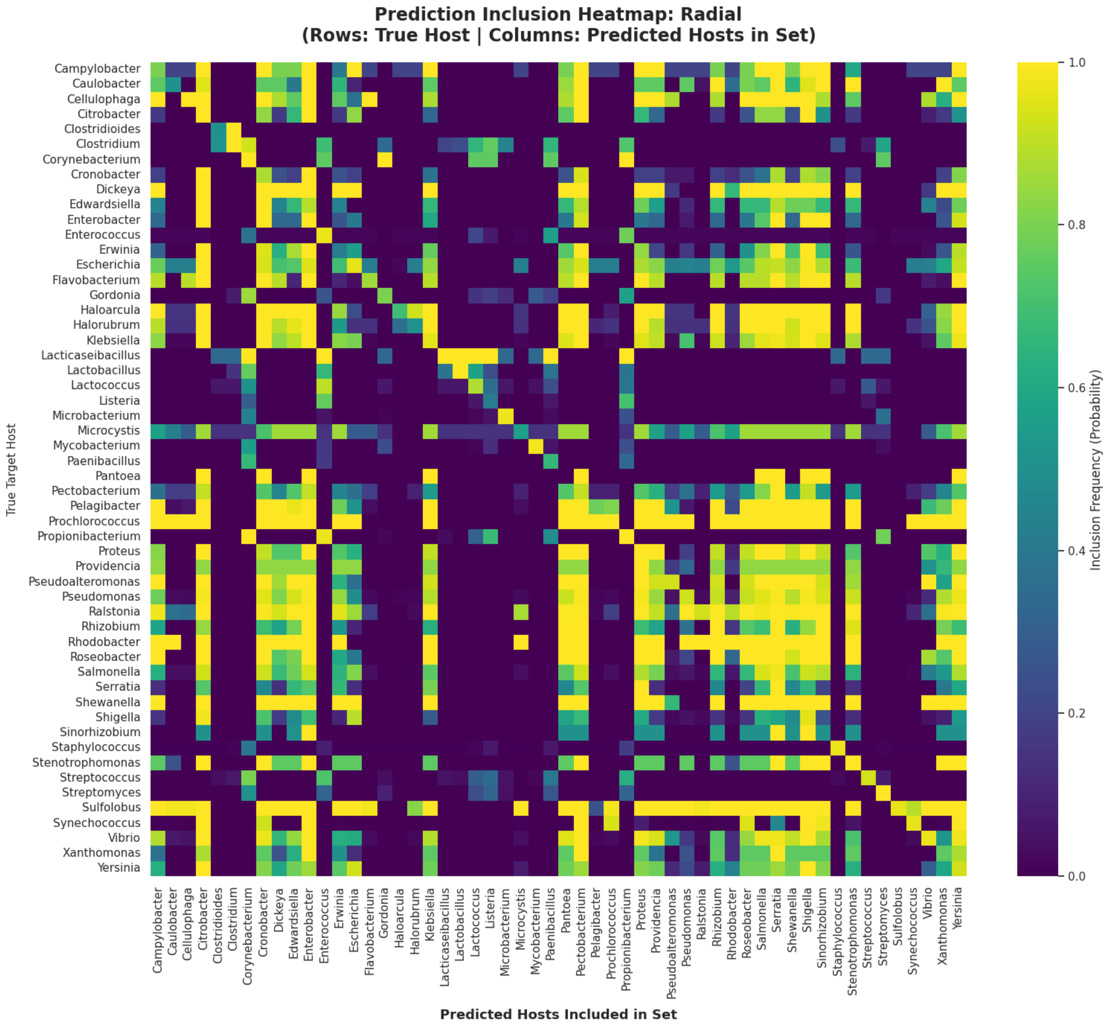

Figure A.8: Prediction set inclusion frequency heatmap for the Radial envelope across all 54 true hosts (rows) vs. predicted candidate hosts (columns).

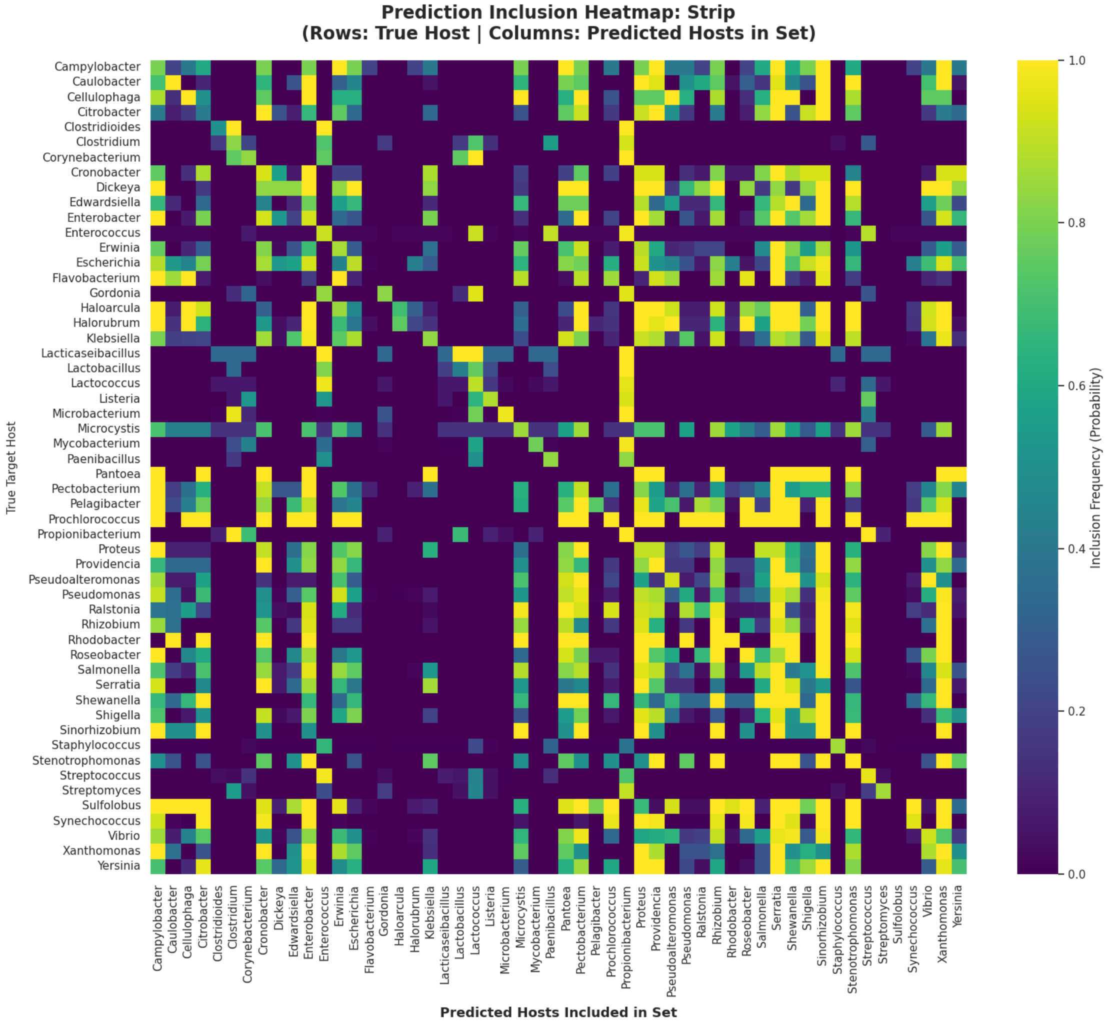

Figure A.9: Prediction set inclusion frequency heatmap for the Strip envelope across all 54 true hosts (rows) vs. predicted candidate hosts (columns).

A.4 Phage Host Prediction Clustermaps

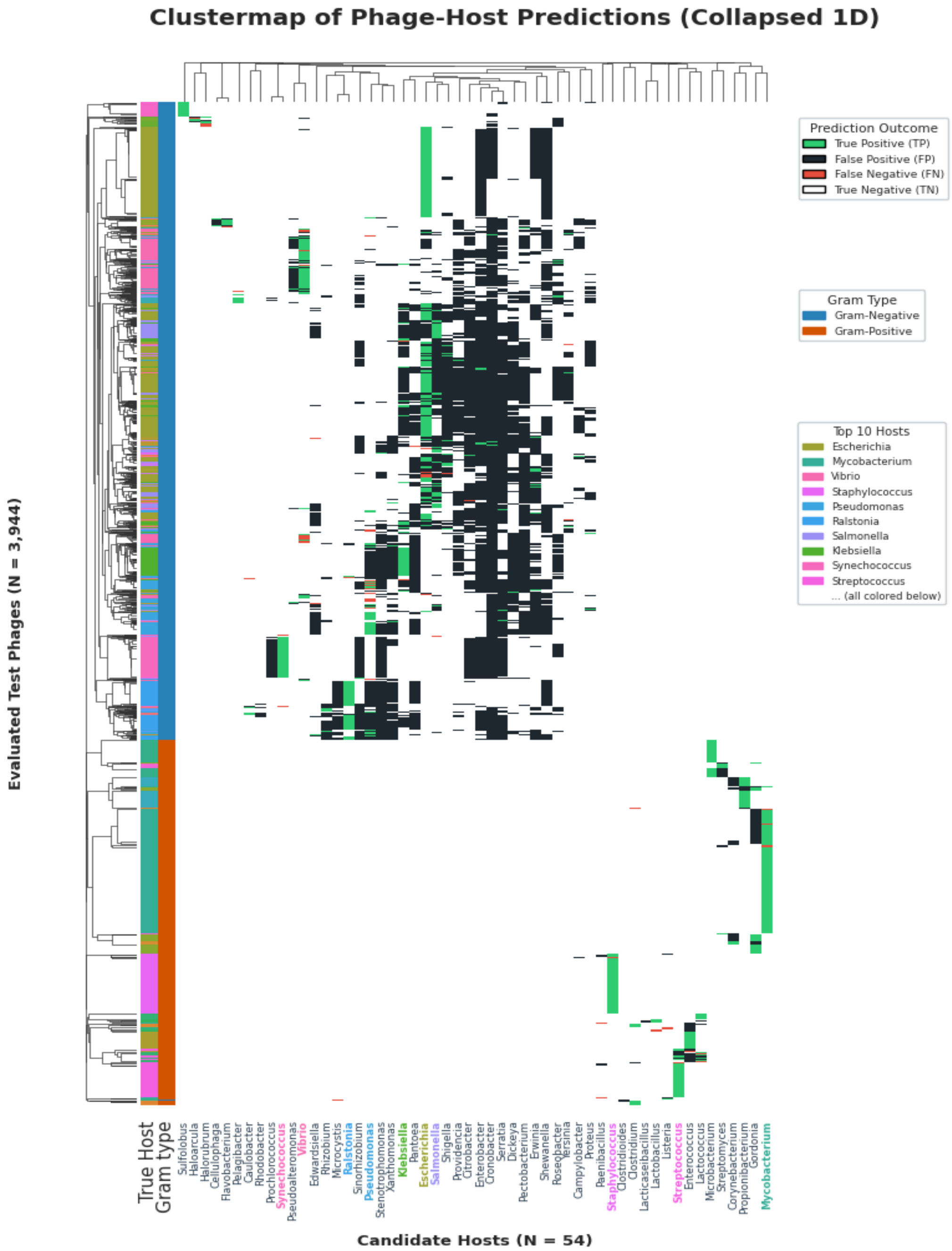

Figure A.10: **Clustermap of Predicted Phage Host Interactions (Radial Envelope).** Binary prediction matrix across $N_{\text{test}} = 3{,}944$ test phages and 54 host genera for the Radial envelope ($M = 100, \kappa = 4.0$).

Figure A.11: **Clustermap of Predicted Phage Host Interactions (Strip Envelope).** Binary prediction matrix across $N_{\text{test}} = 3{,}944$ test phages and 54 host genera for the Strip envelope ($N_B = 10, w = 2$).